

Defeating Physical Intrusion Detection Alarm Wires

Bill Graydon

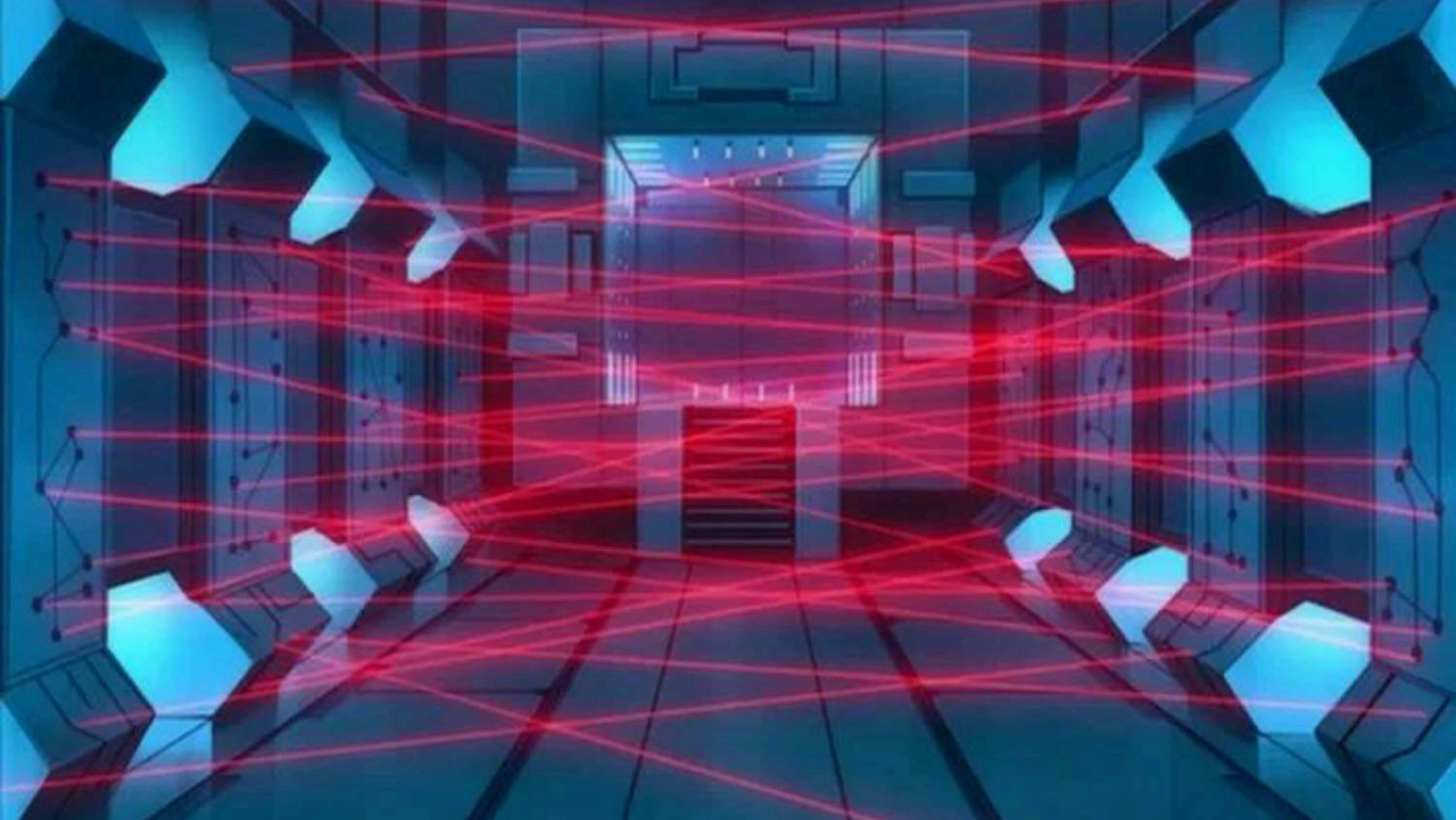


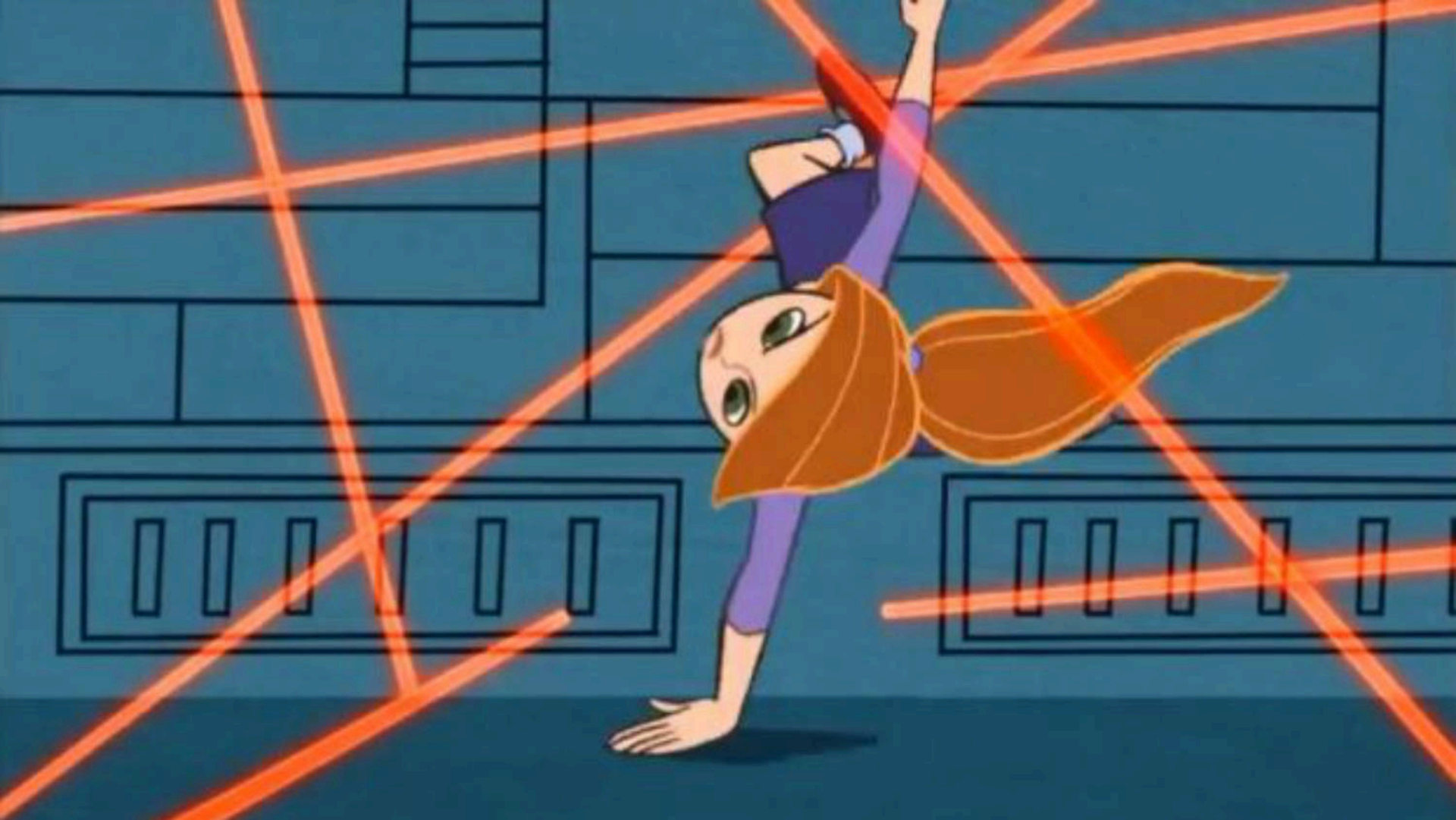
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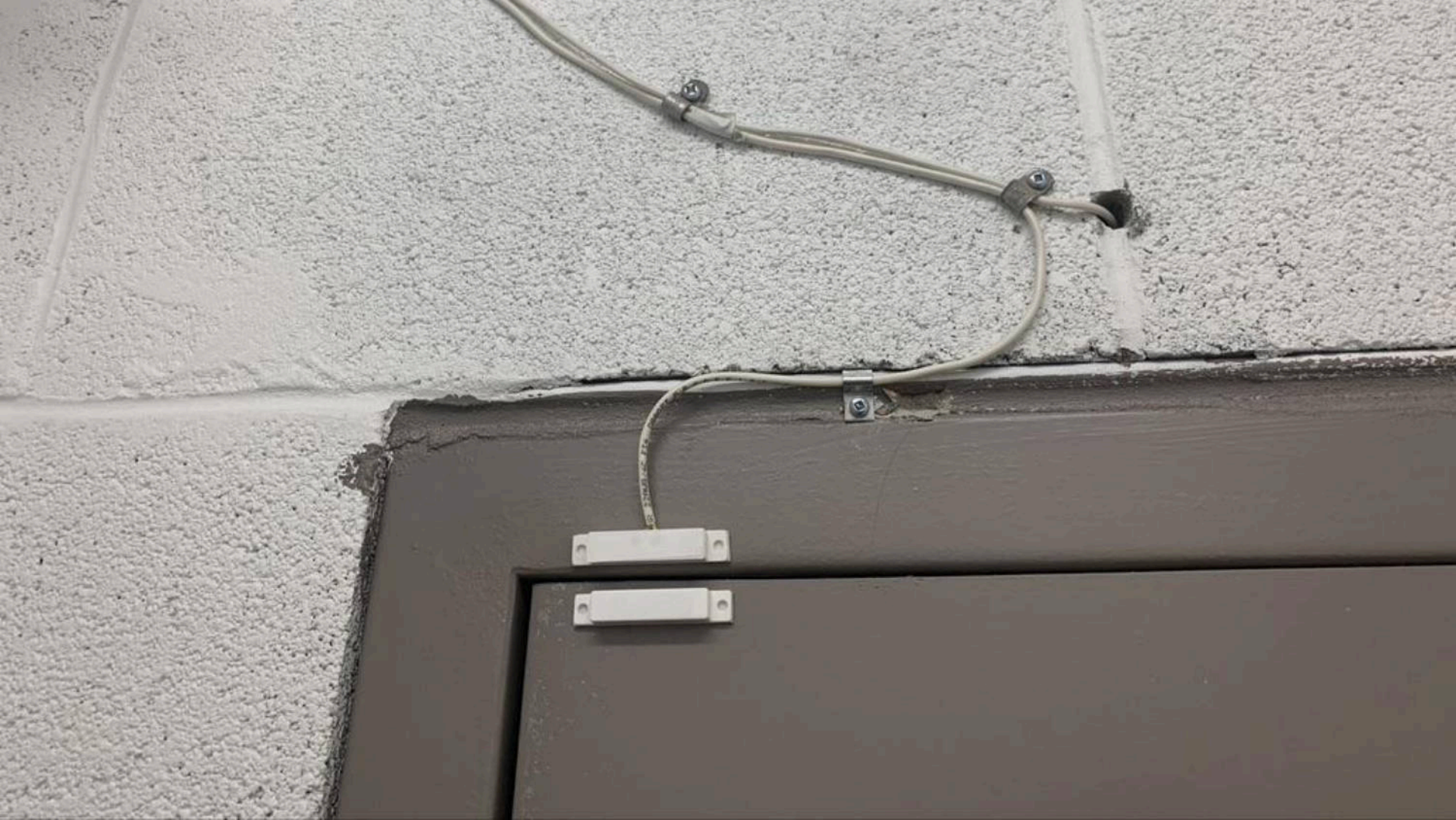
b.graydon@ggrsecurity.com

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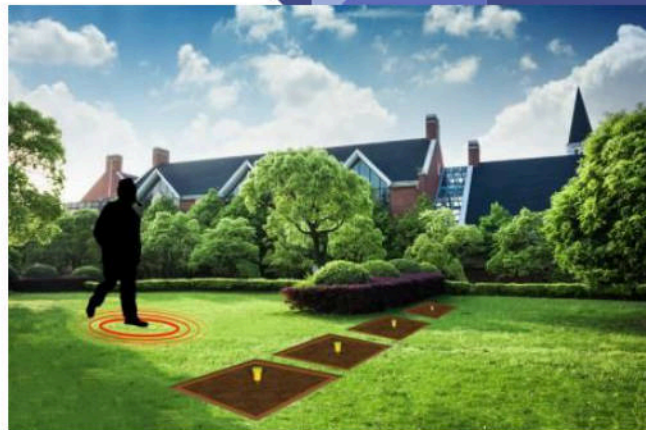
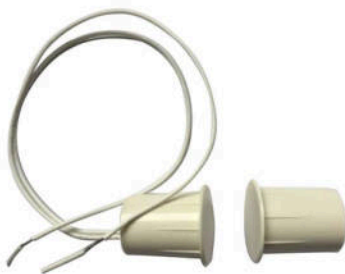


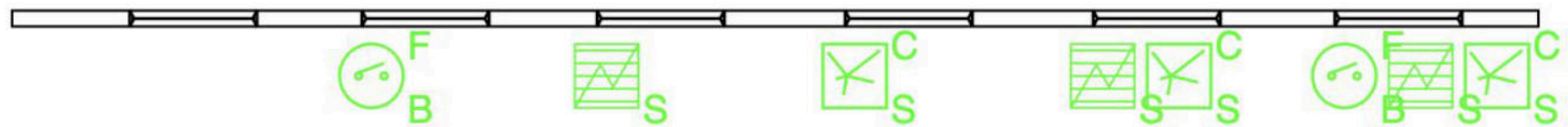
This is a talk about sensor communication wires.

- Alarm systems and access control first
- Defeating line supervision and end-of-line resistors
 - Surrogate Resistor
 - Voltage Regulation
- Defenses

Go try it in the Lock Bypass Village!

https://www.bypassvillage.org/games/alarm_wire/





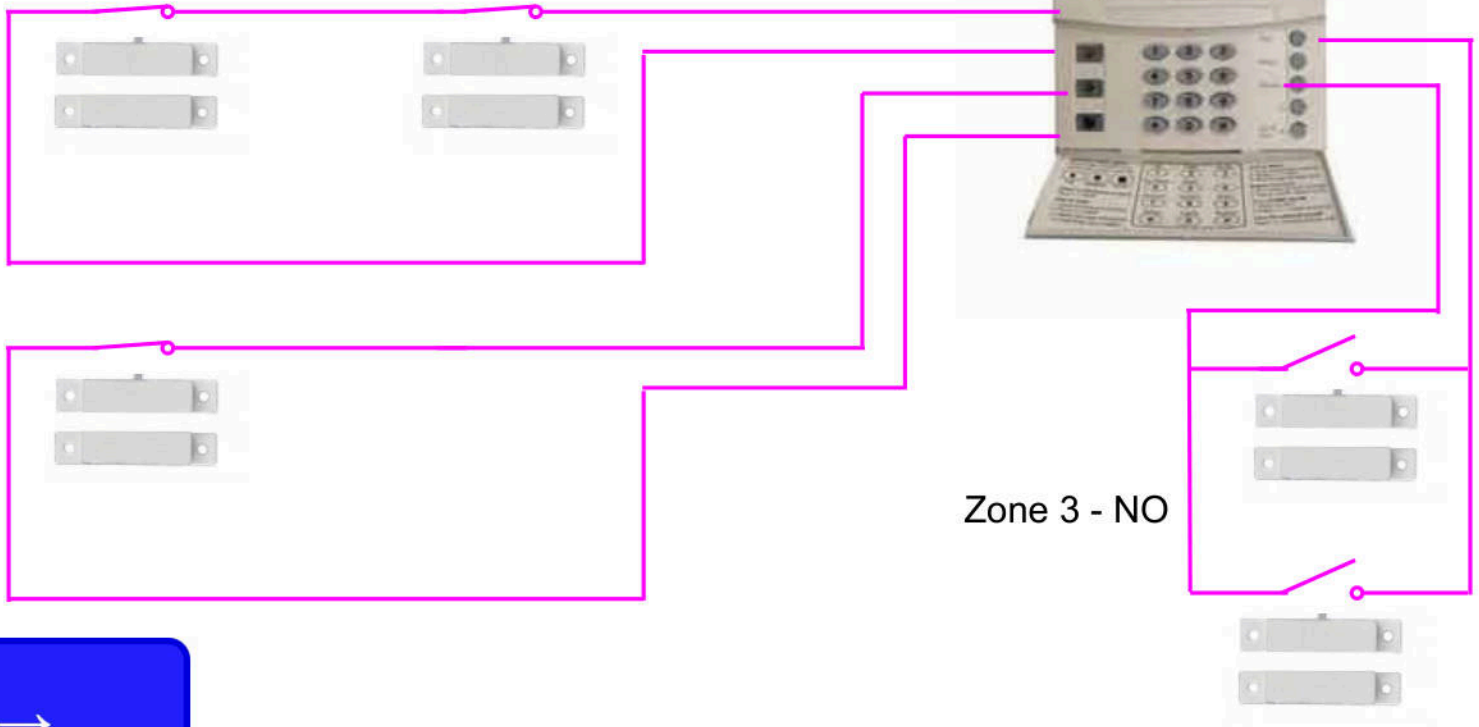
Alarm Zones

Zone 1 - NC

Zone 2 - NC

Zone 3 - NO

Demo →





NIGHT BELL

DAY BELL

STOP

OUTBREAK ALERT

AUTOMATIC CAUTION DOOR

Service Animals Welcome

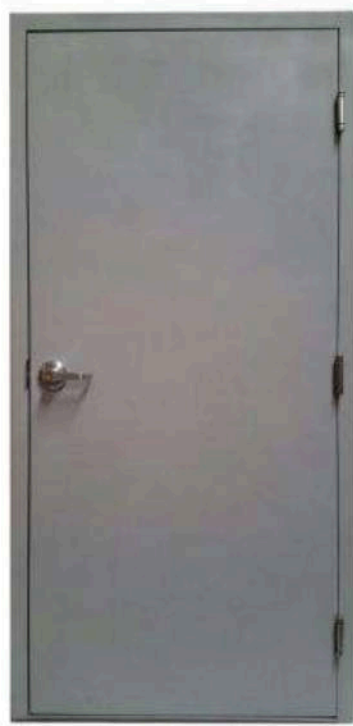
NOTICE

740

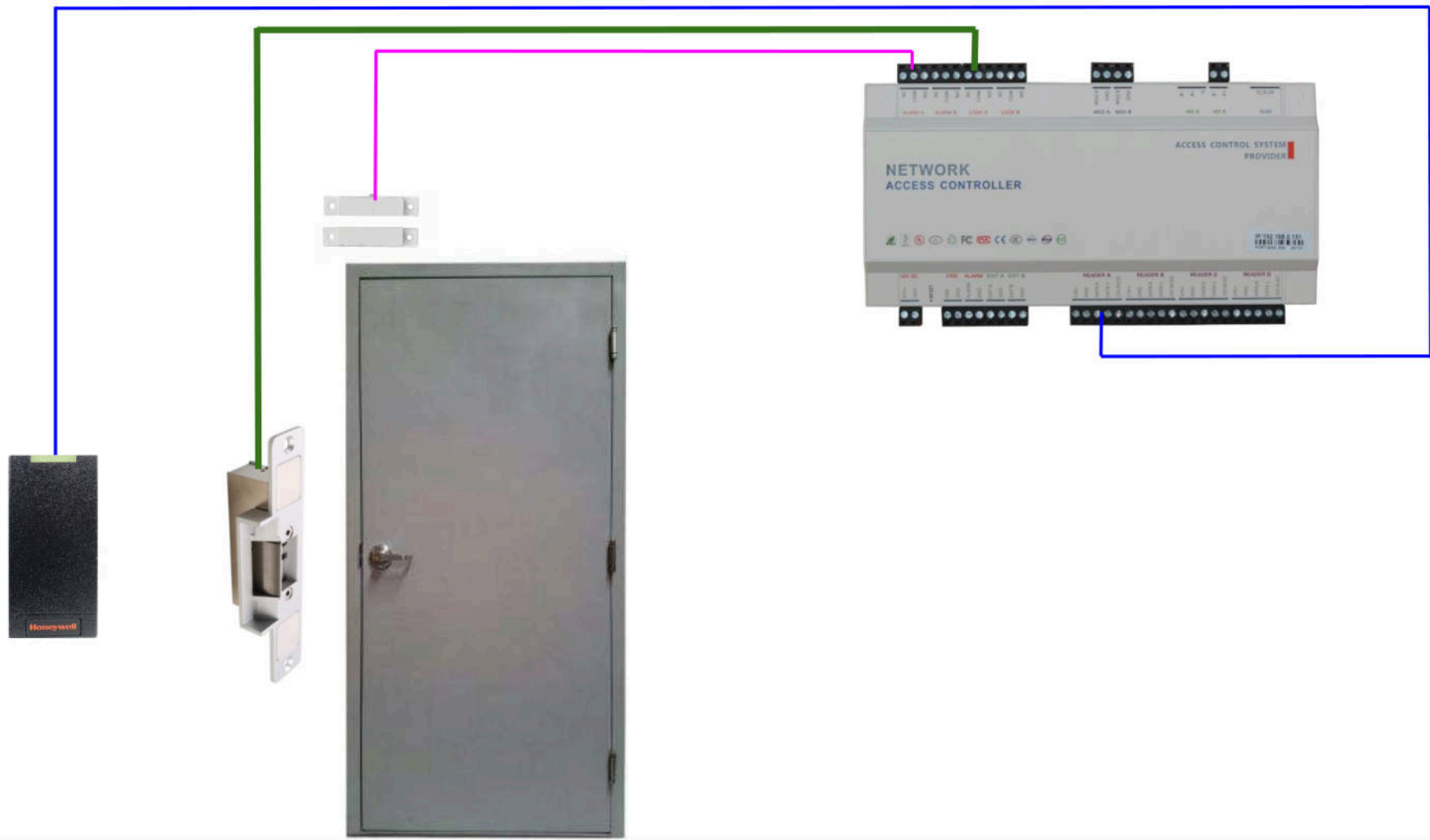
FIRE PANEL

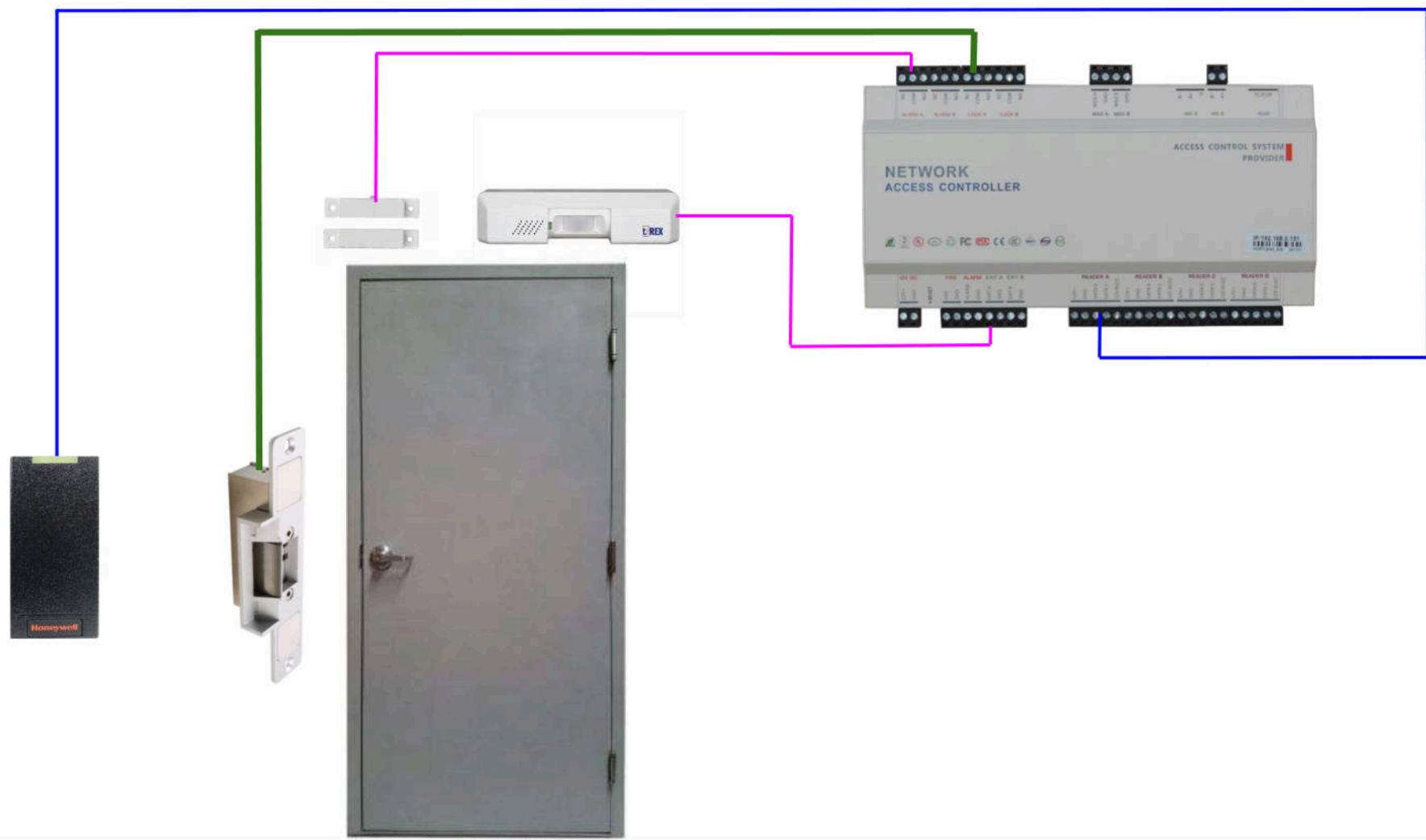
FIRE PANEL RESET
LOCATED IN B-2

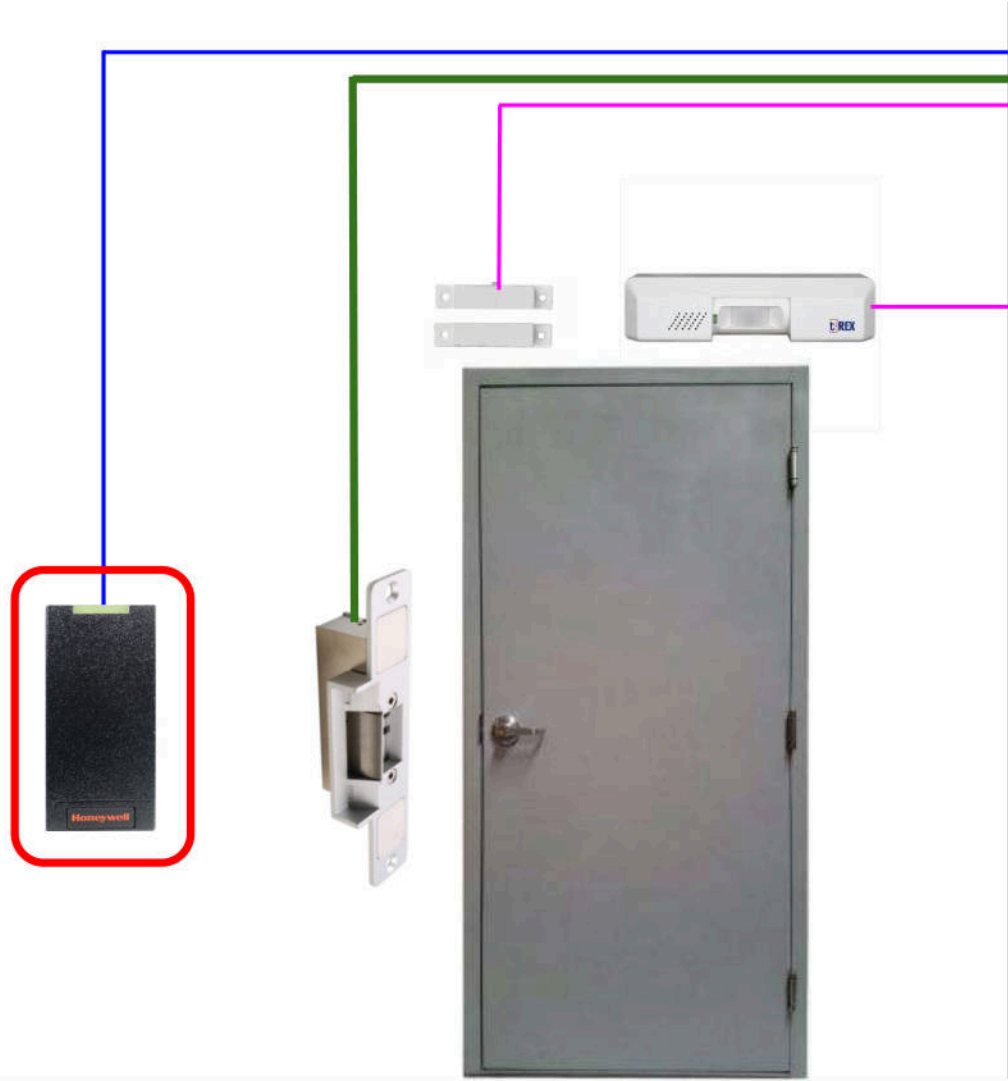


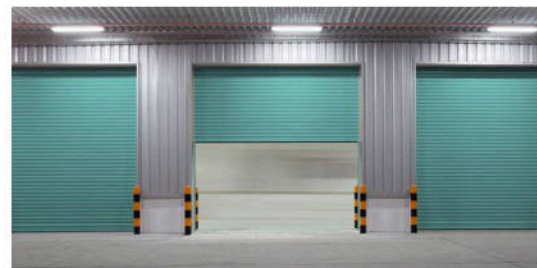
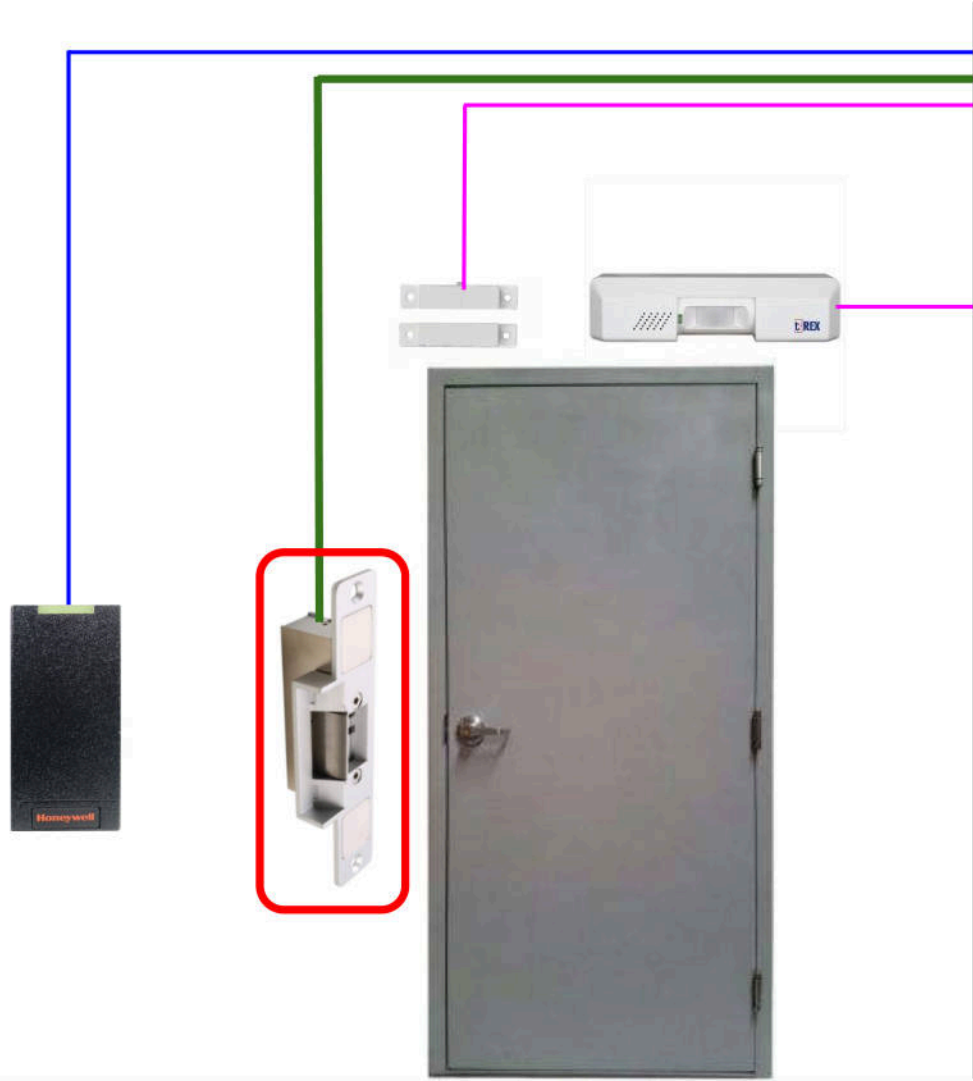


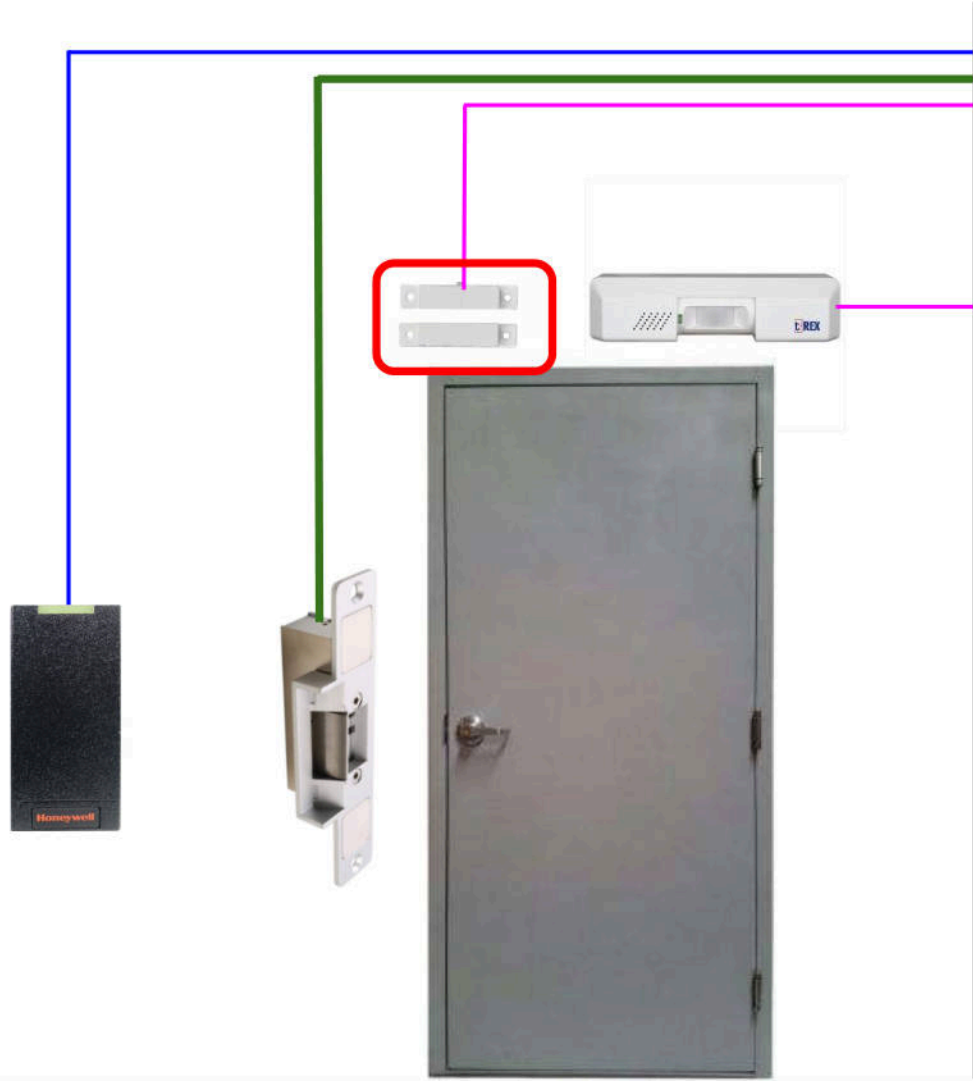








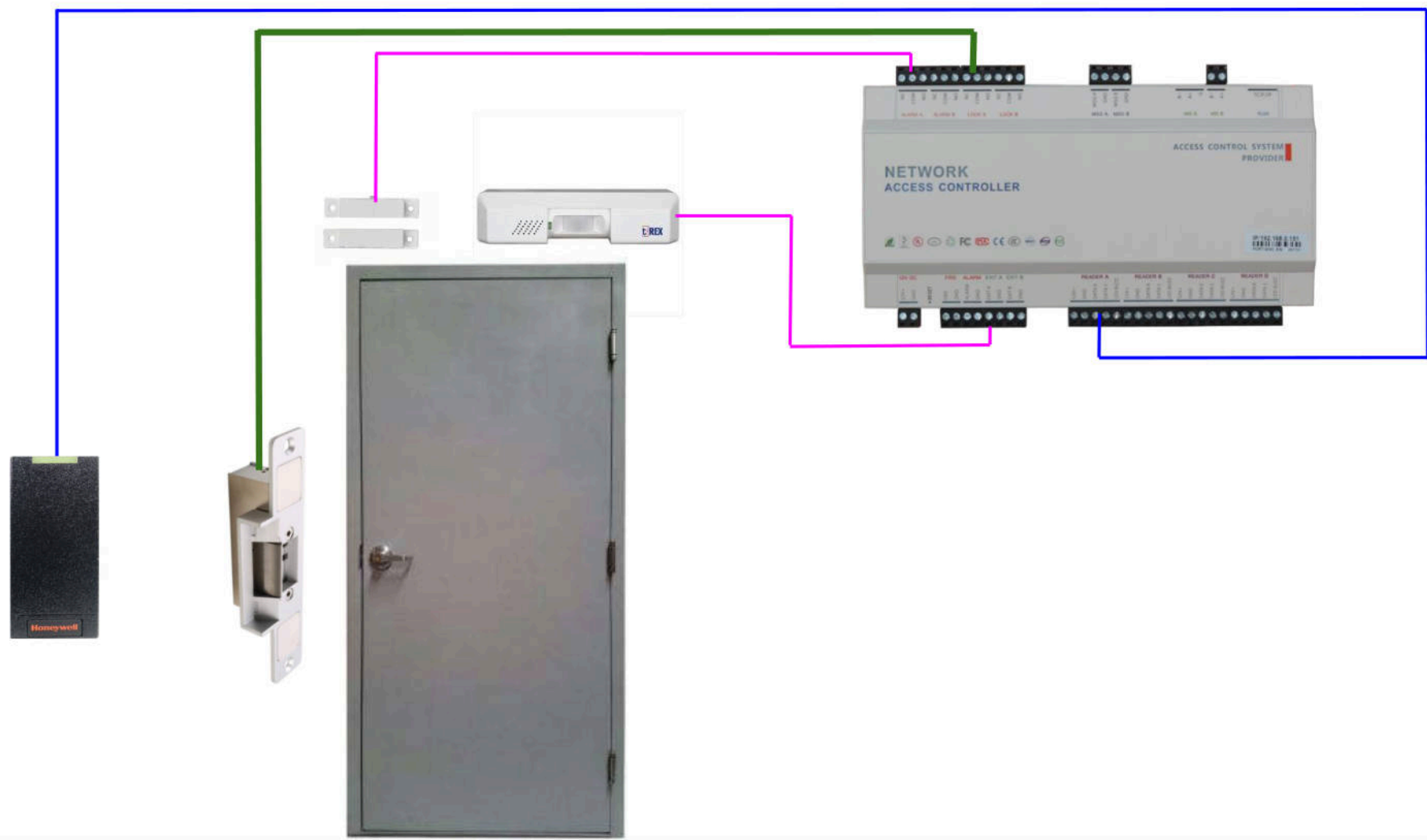


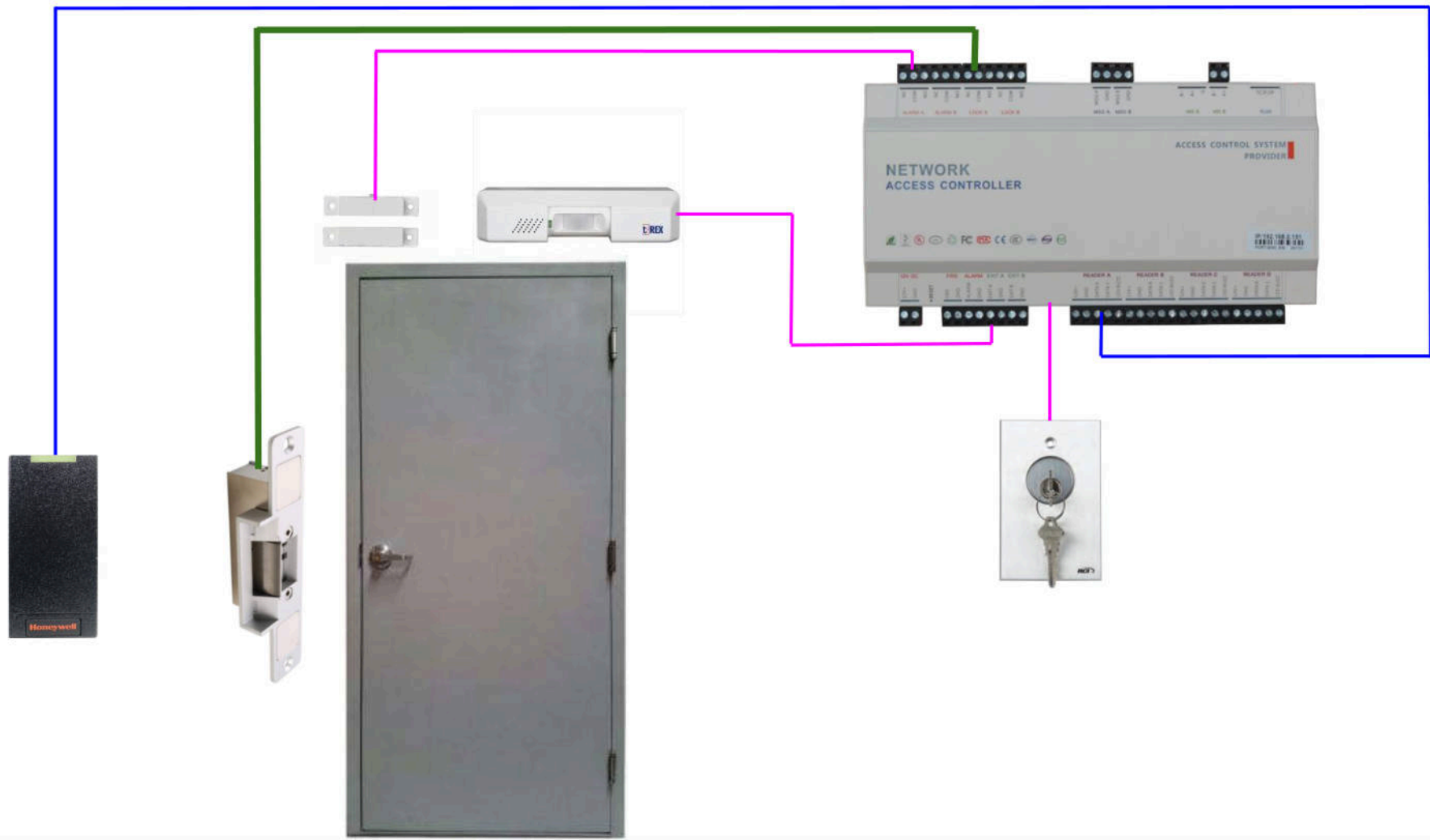








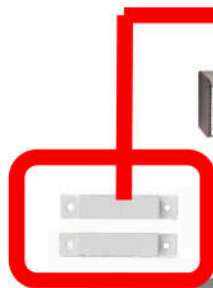


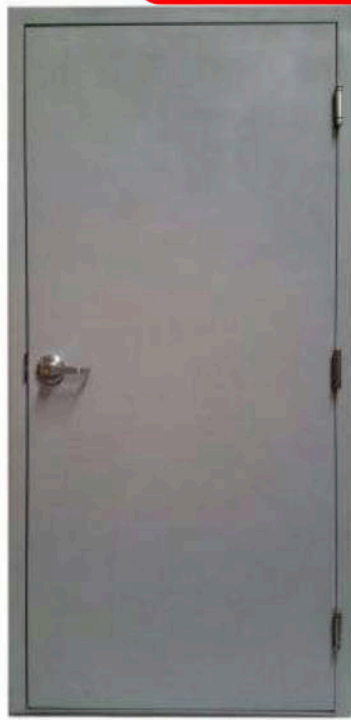


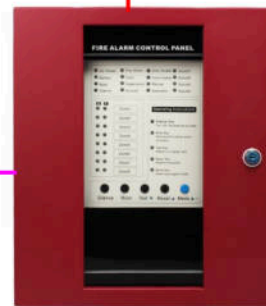
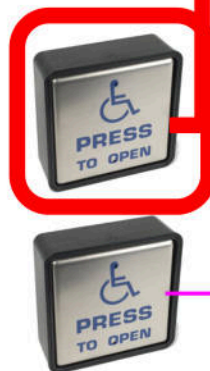






















F/A
DOOR CONTACTS

SEC JB





RD 6E Y BK

BELL PHONE
ONLY



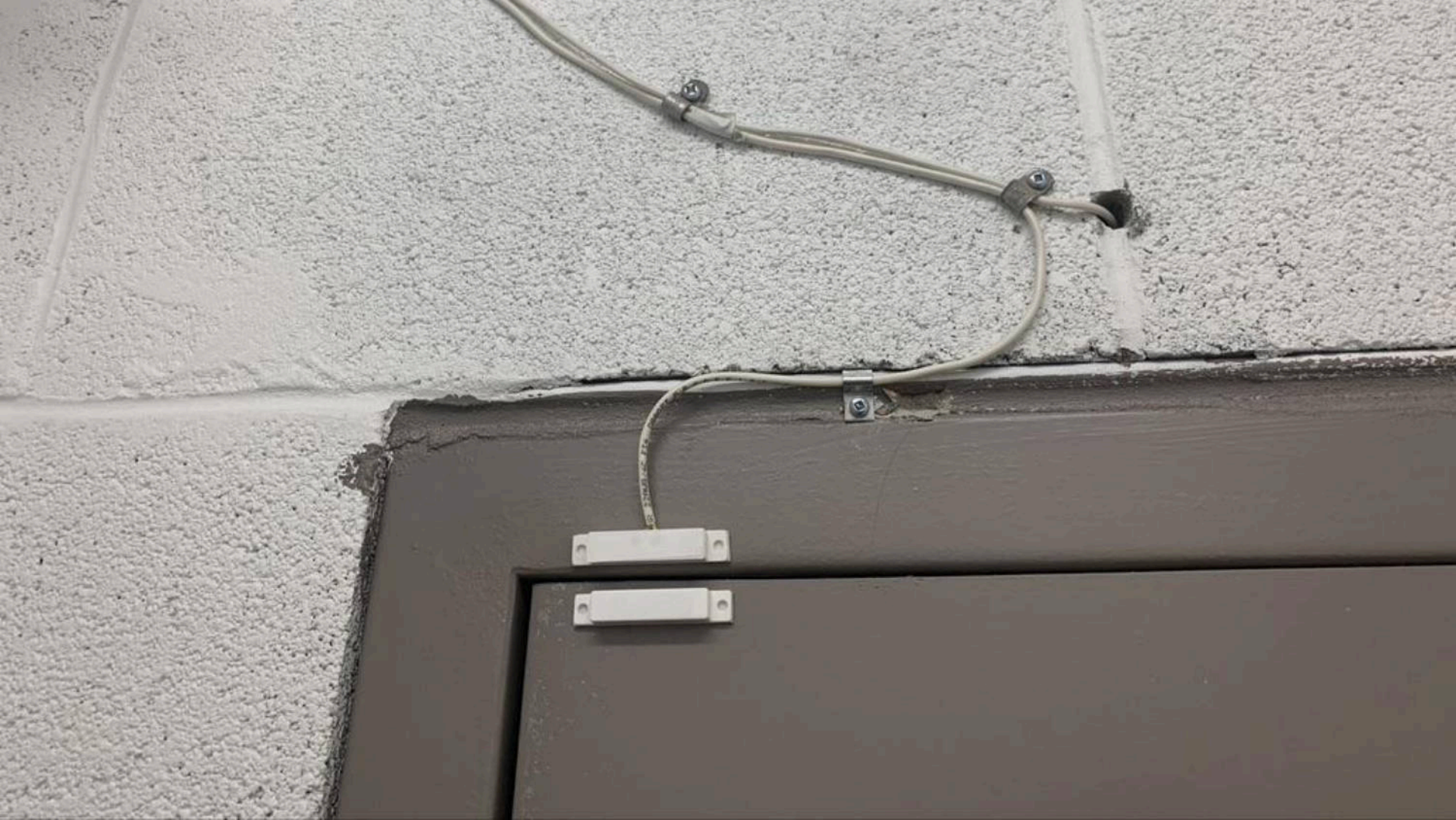
SECURITY SYSTEM RM.













PKS
DOOR KING

Red Team Attacks - Binary Sensor Comms Lines - Normally Closed



Switch closed - controller sees low impedance - **no alarm**

Switch open - controller sees high impedance - **alarm**

Switch open, line jumpered - controller sees low impedance - **no alarm**

Demo →

Red Team Attacks - Binary Sensor Comms Lines - Normally Open



Switch open - controller sees high impedance - **no alarm**



Switch closed - controller sees low impedance - **alarm**



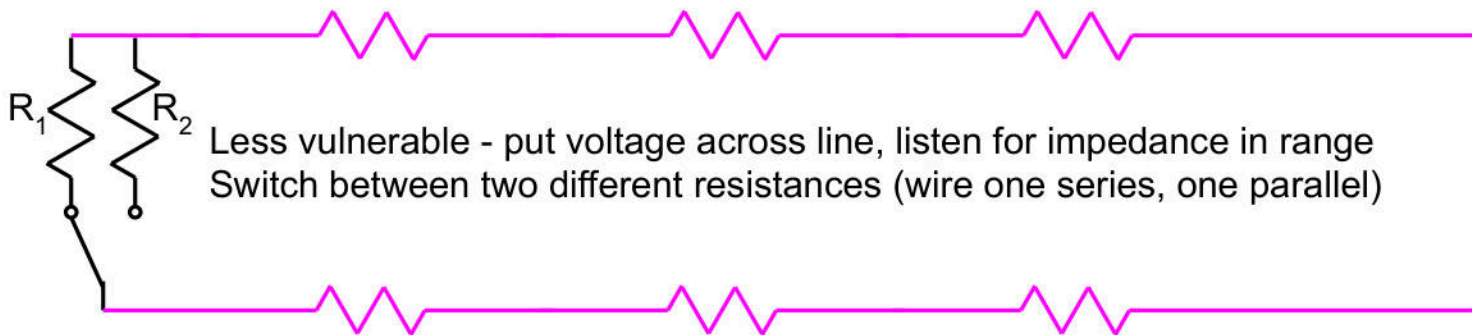
Switch closed, line cut - controller sees high impedance - **no alarm**



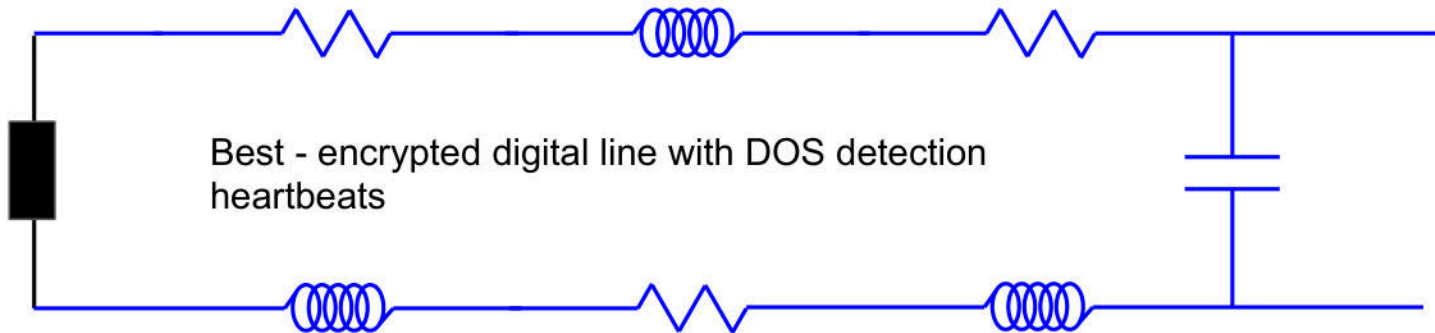
Demo →

Blue Team Defense - Binary Sensor Comms Lines

Vulnerable - put voltage across line, listen for high / low impedance



Best - encrypted digital line with DOS detection heartbeats



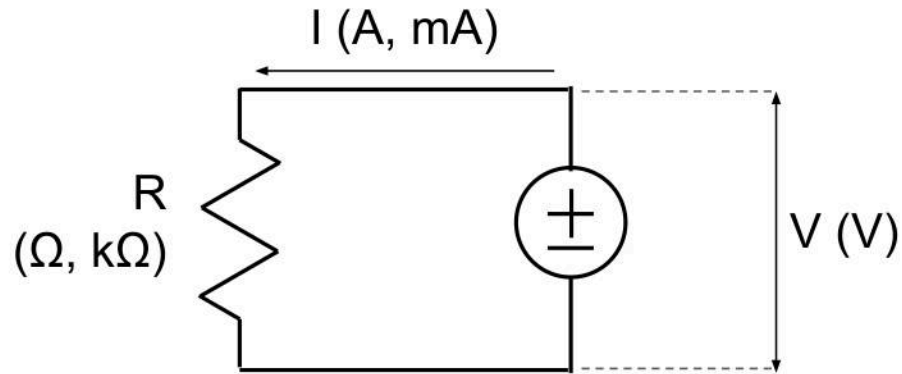
EOLR

End-of-line Resistors

Two Approaches

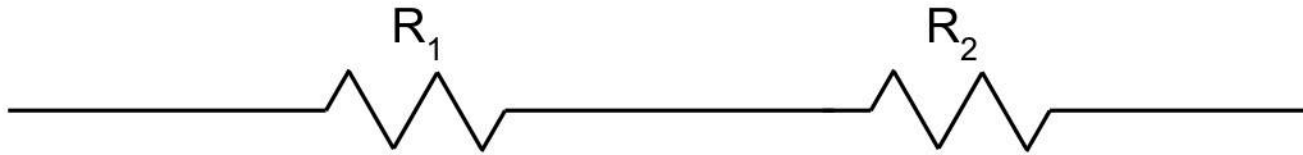
- 1) Surrogate Resistor
- 2) Voltage Regulation

Resistor Review (Slide 1 of 3)



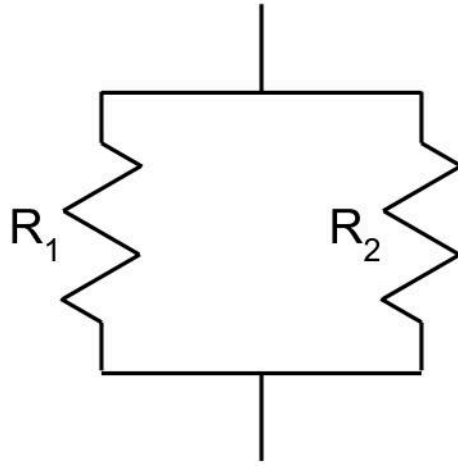
$$R = \frac{V}{I}$$

Resistor Review (Slide 2 of 3)

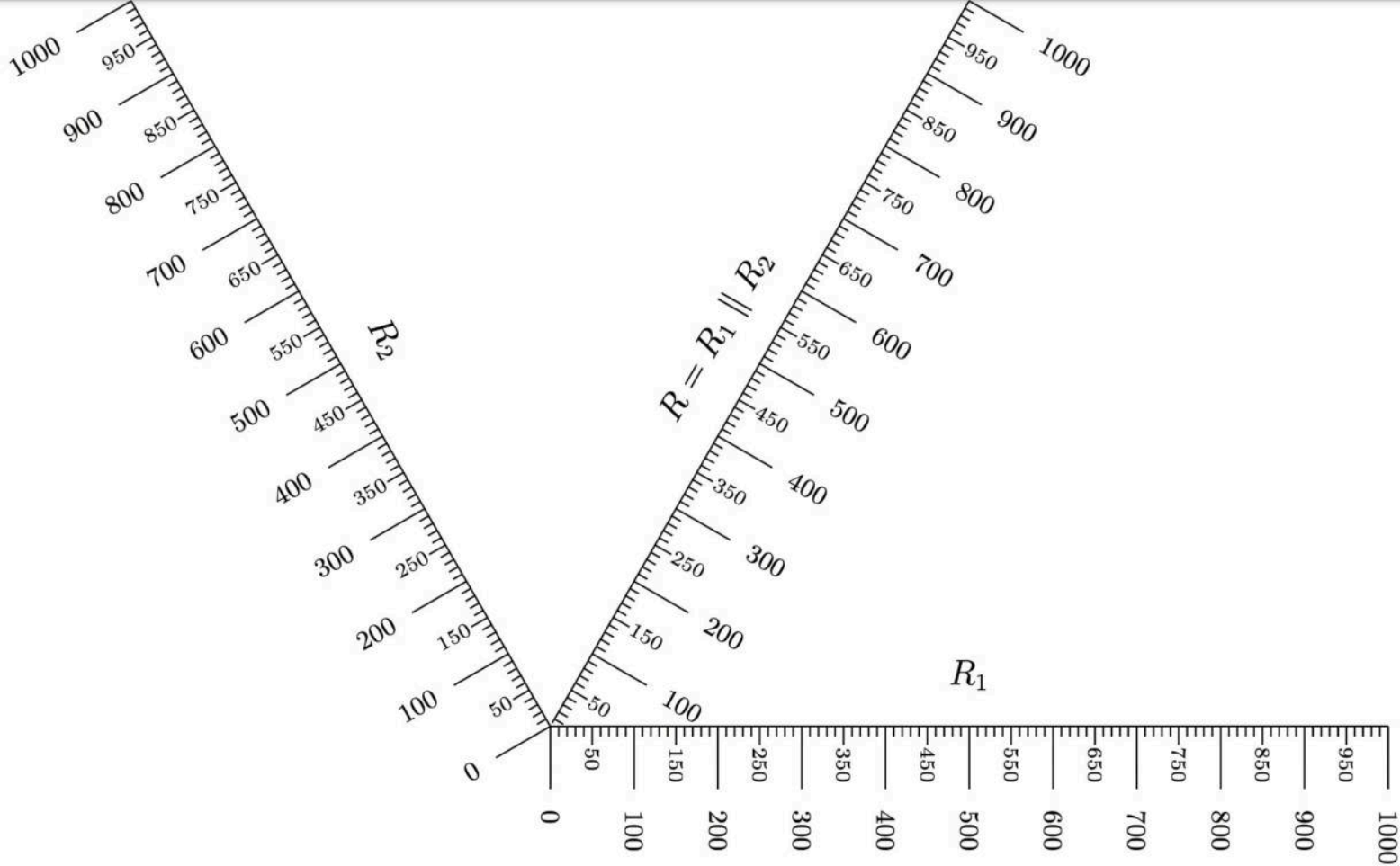


$$R_{\text{equivalent}} = R_1 + R_2$$

Resistor Review (Slide 3 of 3)

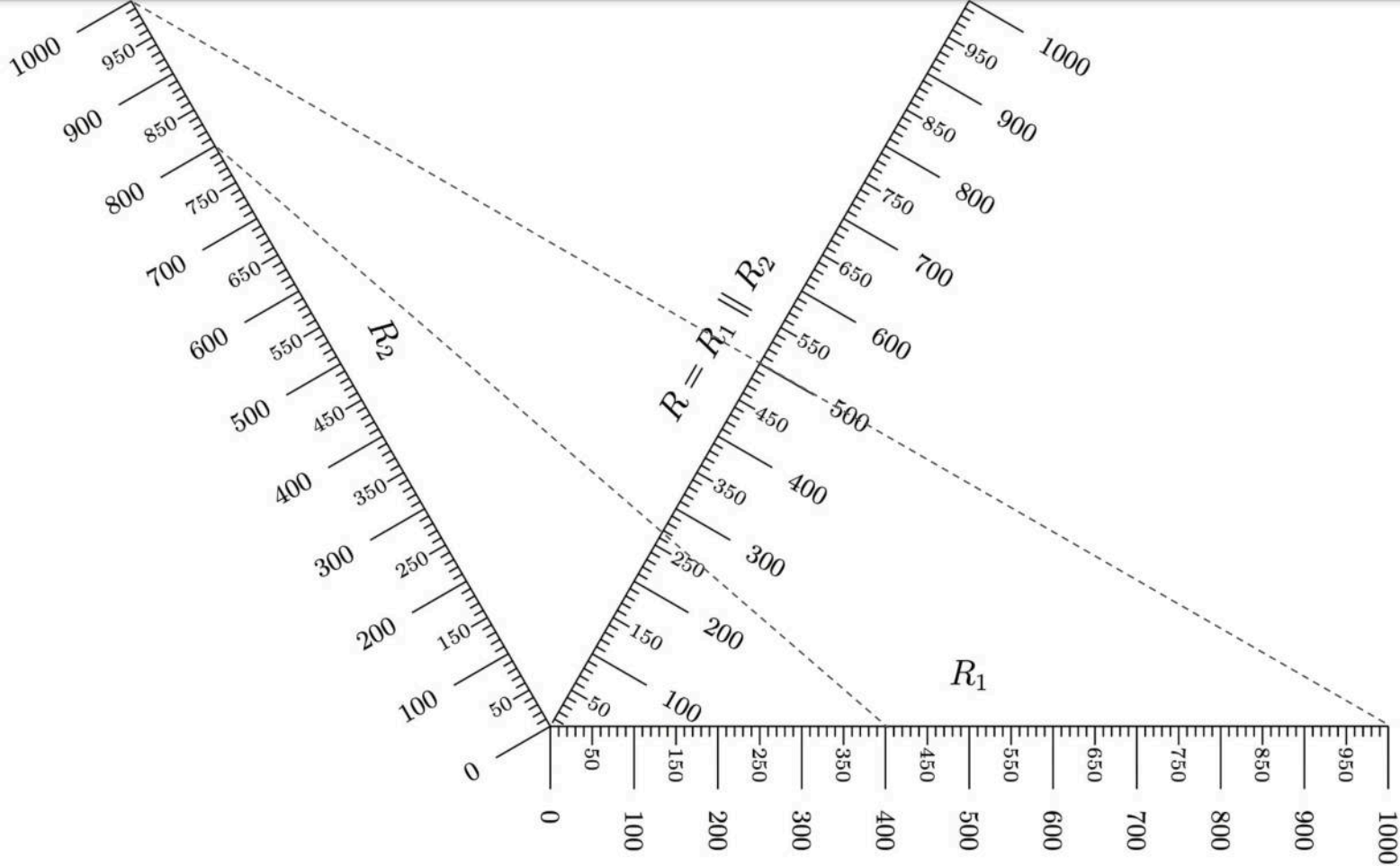


$$\frac{1}{R_{\text{equivalent}}} = \frac{1}{R_1} + \frac{1}{R_2}$$



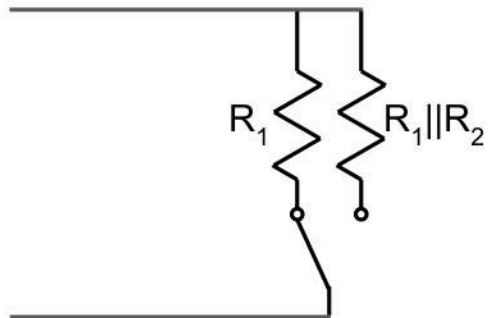
Equivalent Resistance of Two Resistors in Parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

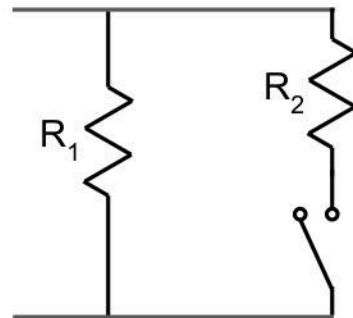


Equivalent Resistance of Two Resistors in Parallel

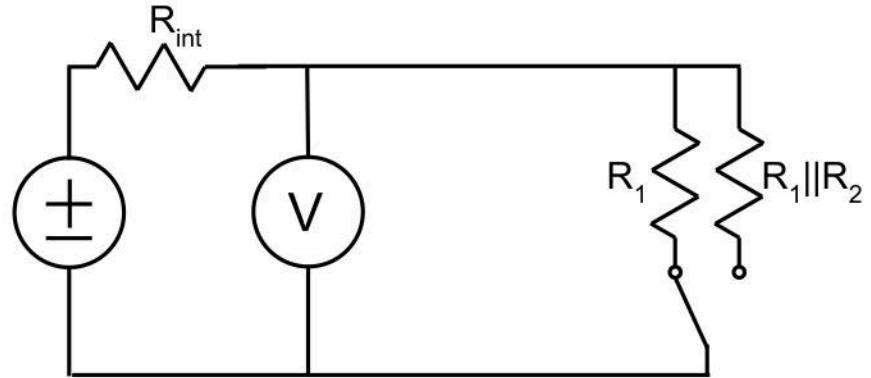
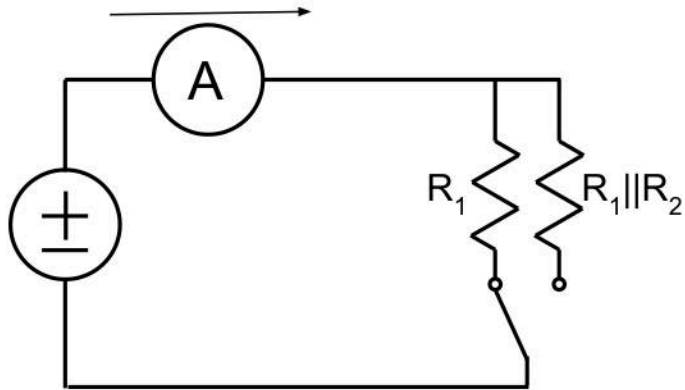
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$



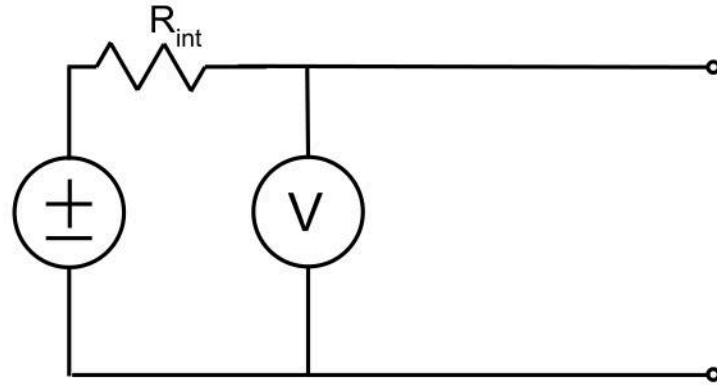
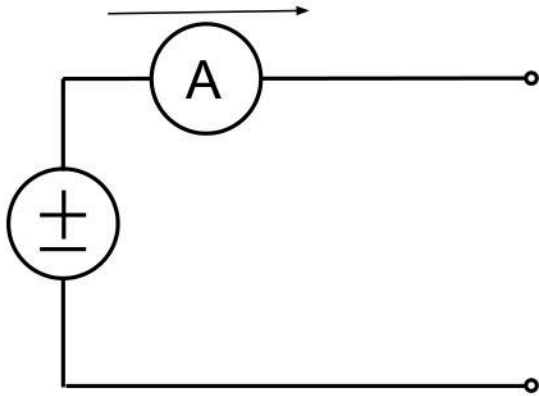
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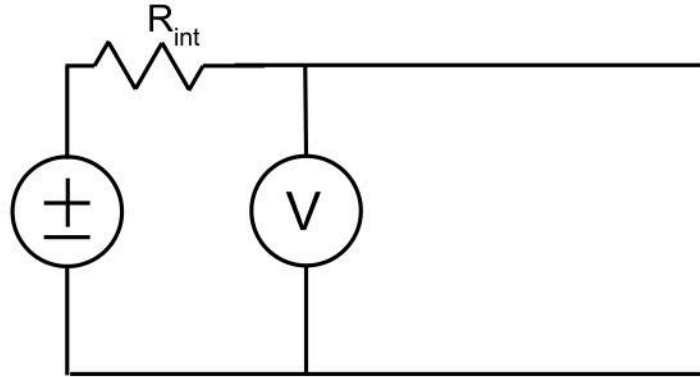
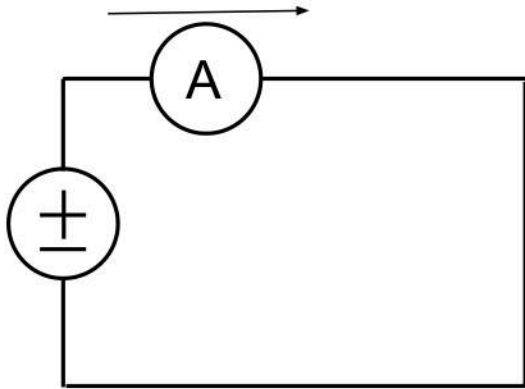
How does the controller measure resistance?



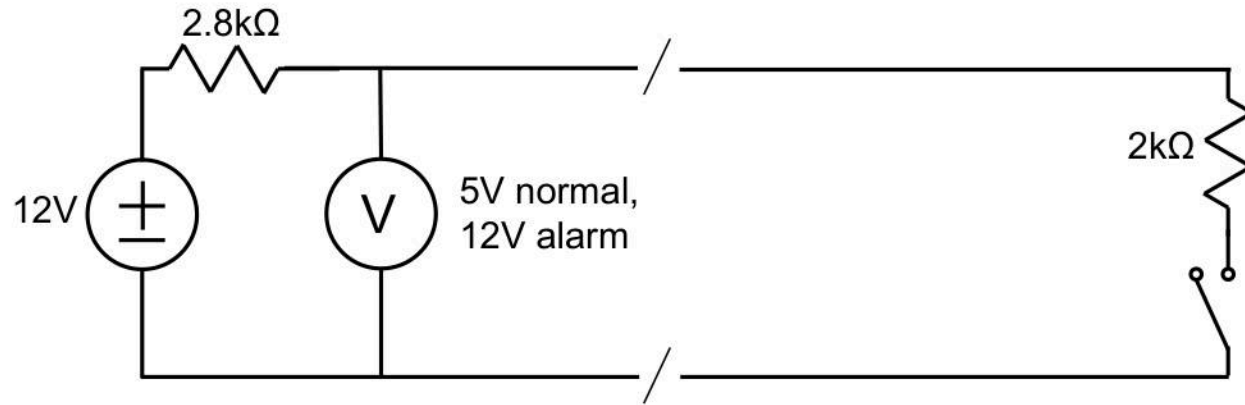
Special case: open circuit (line cut)

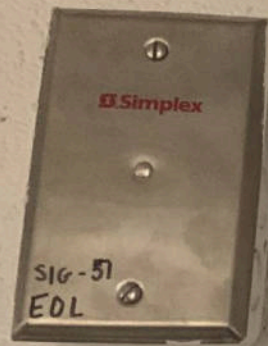


Special case: short circuit (line jumpered)

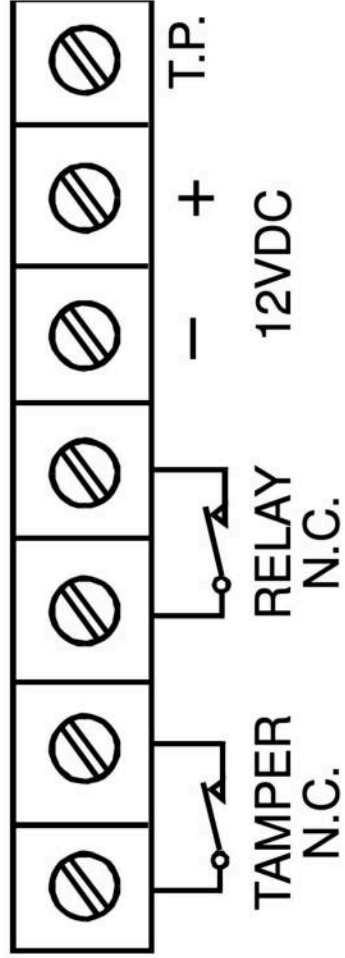


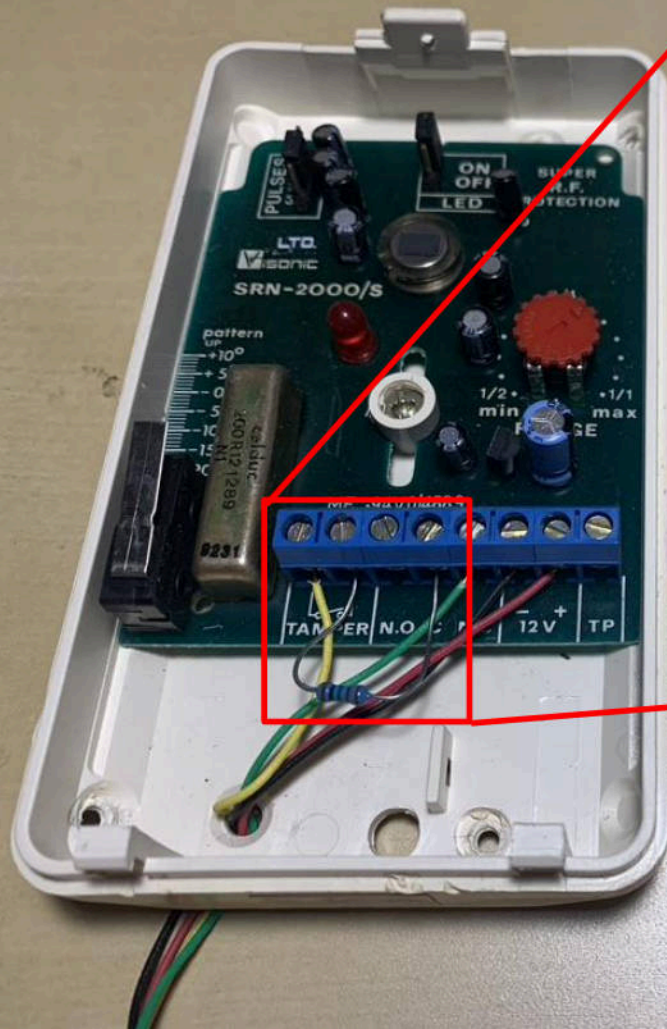
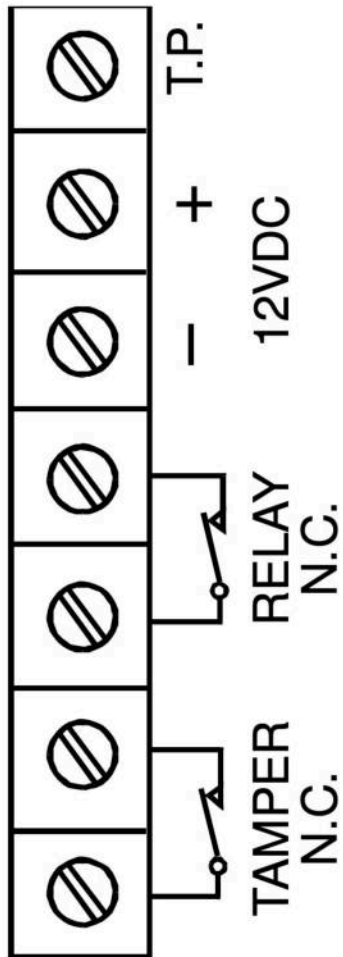
E.g. Honeywell



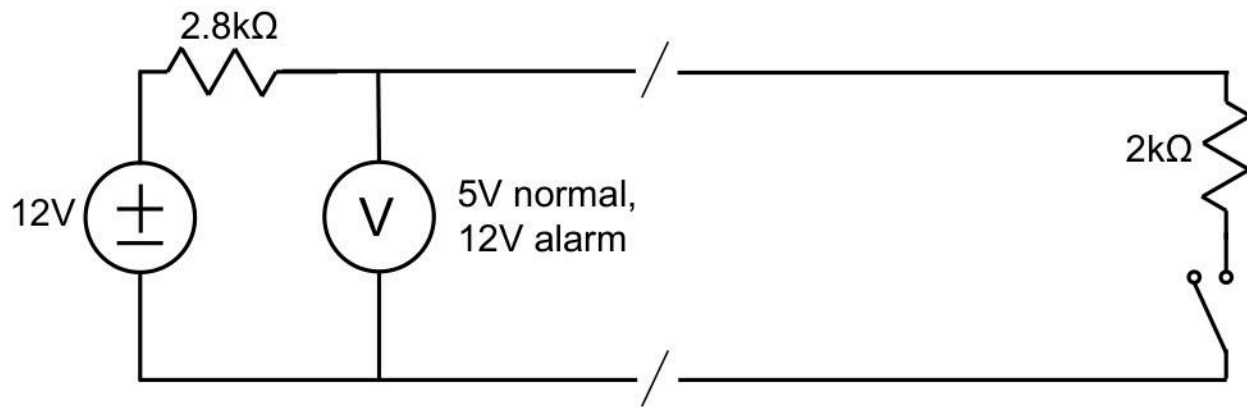




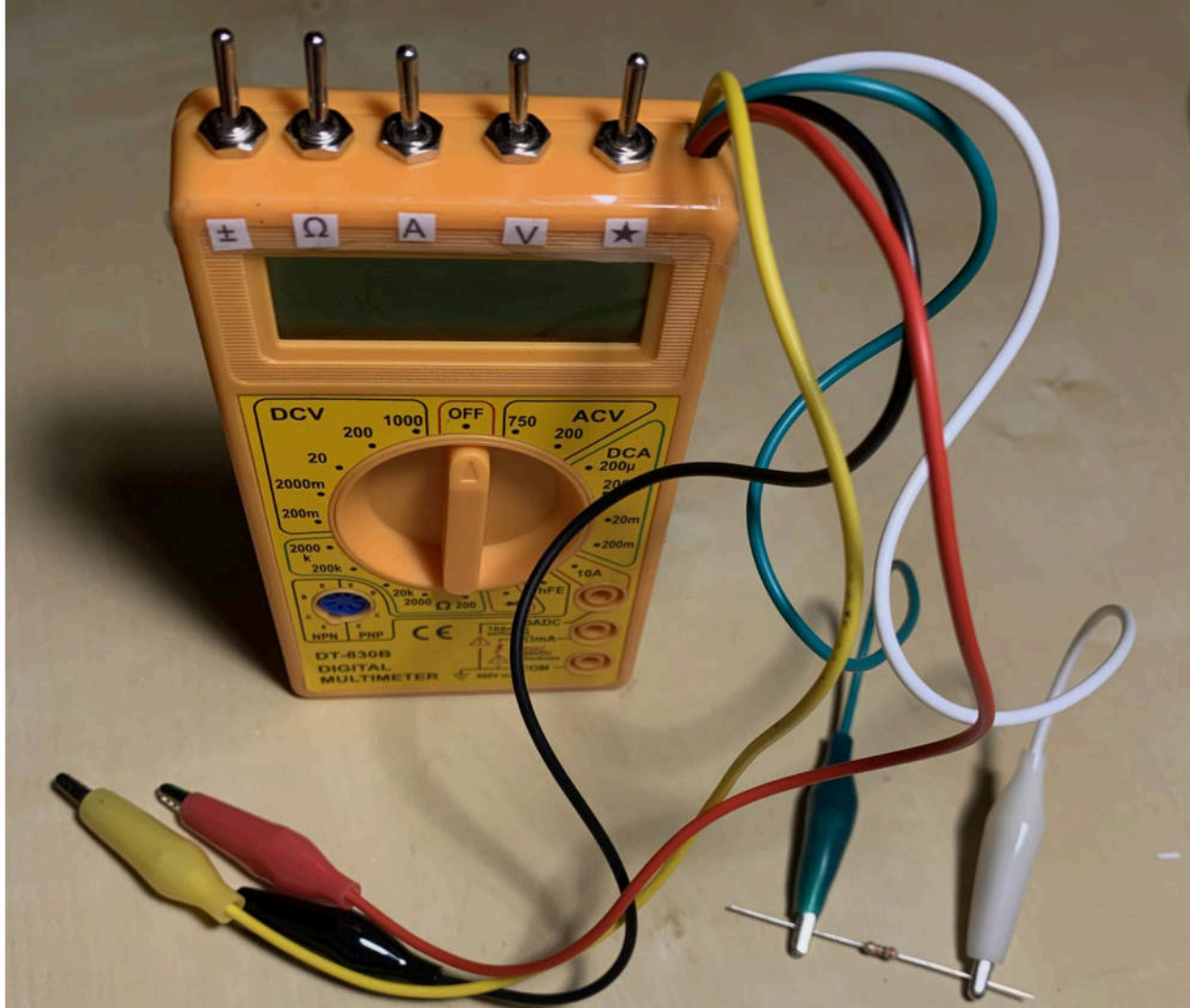




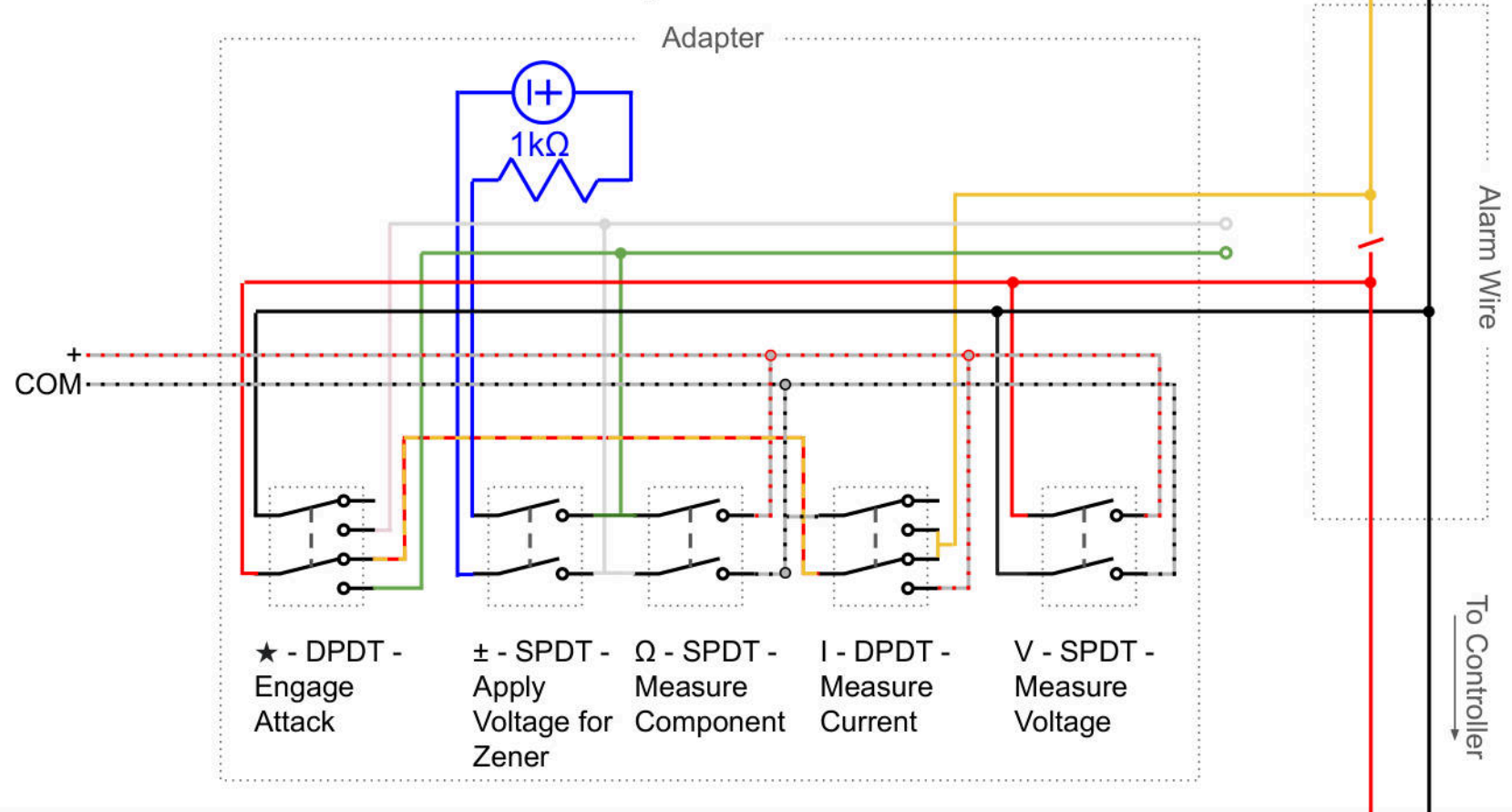
Attacking the EOLRs

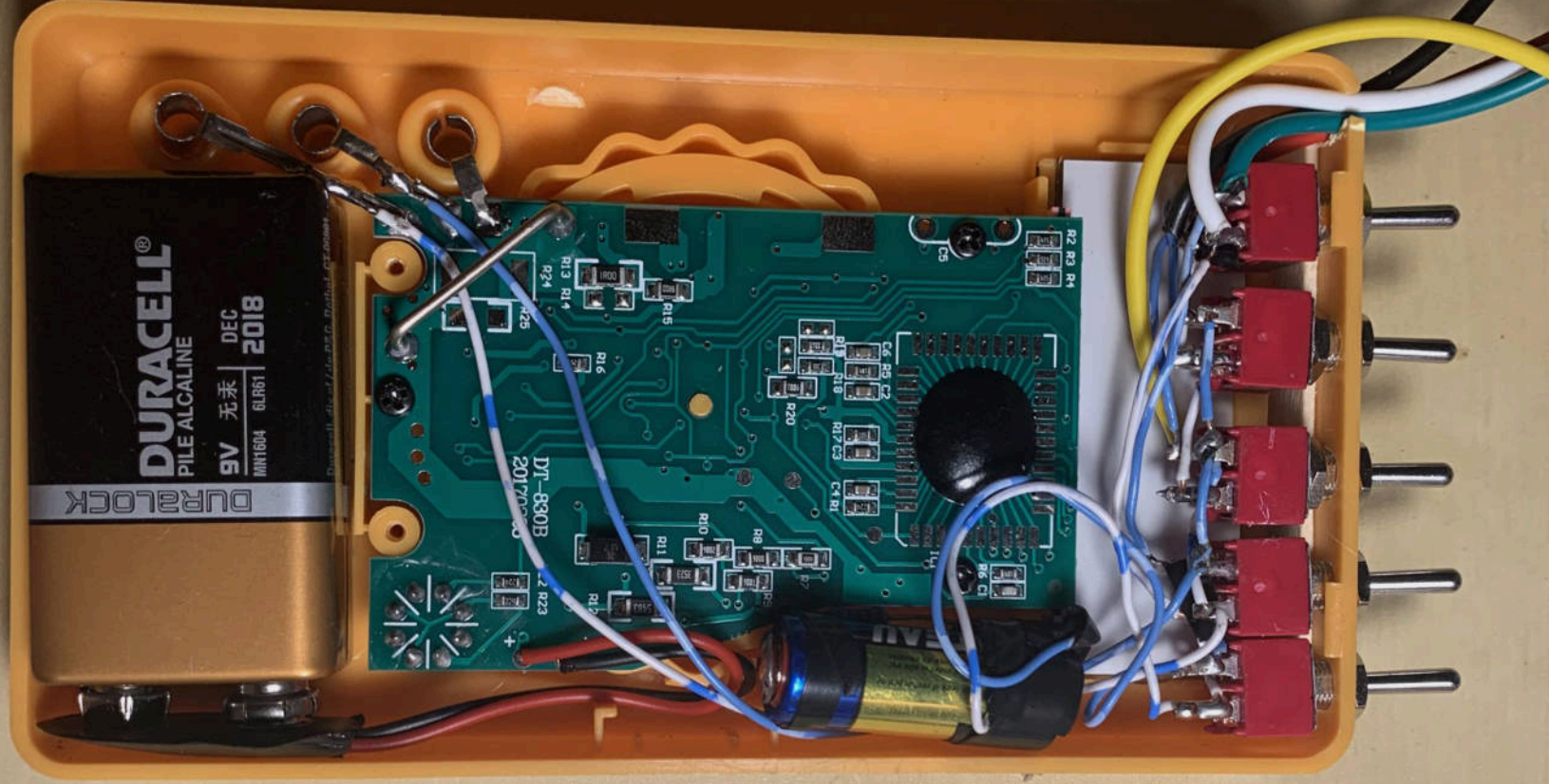


Demo →



Schematic - Multimeter Adapter





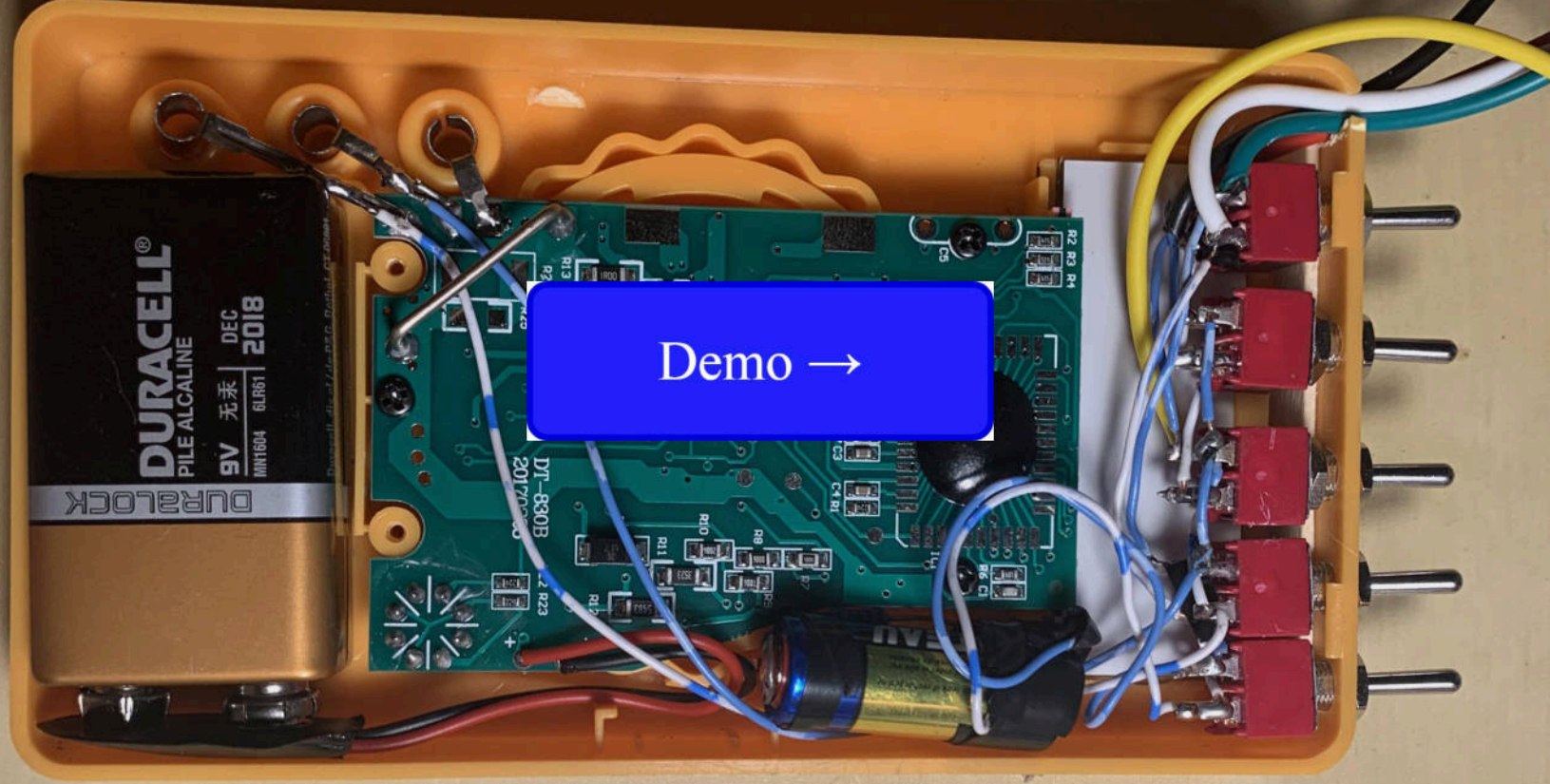
DURALOCK

DURACELL®

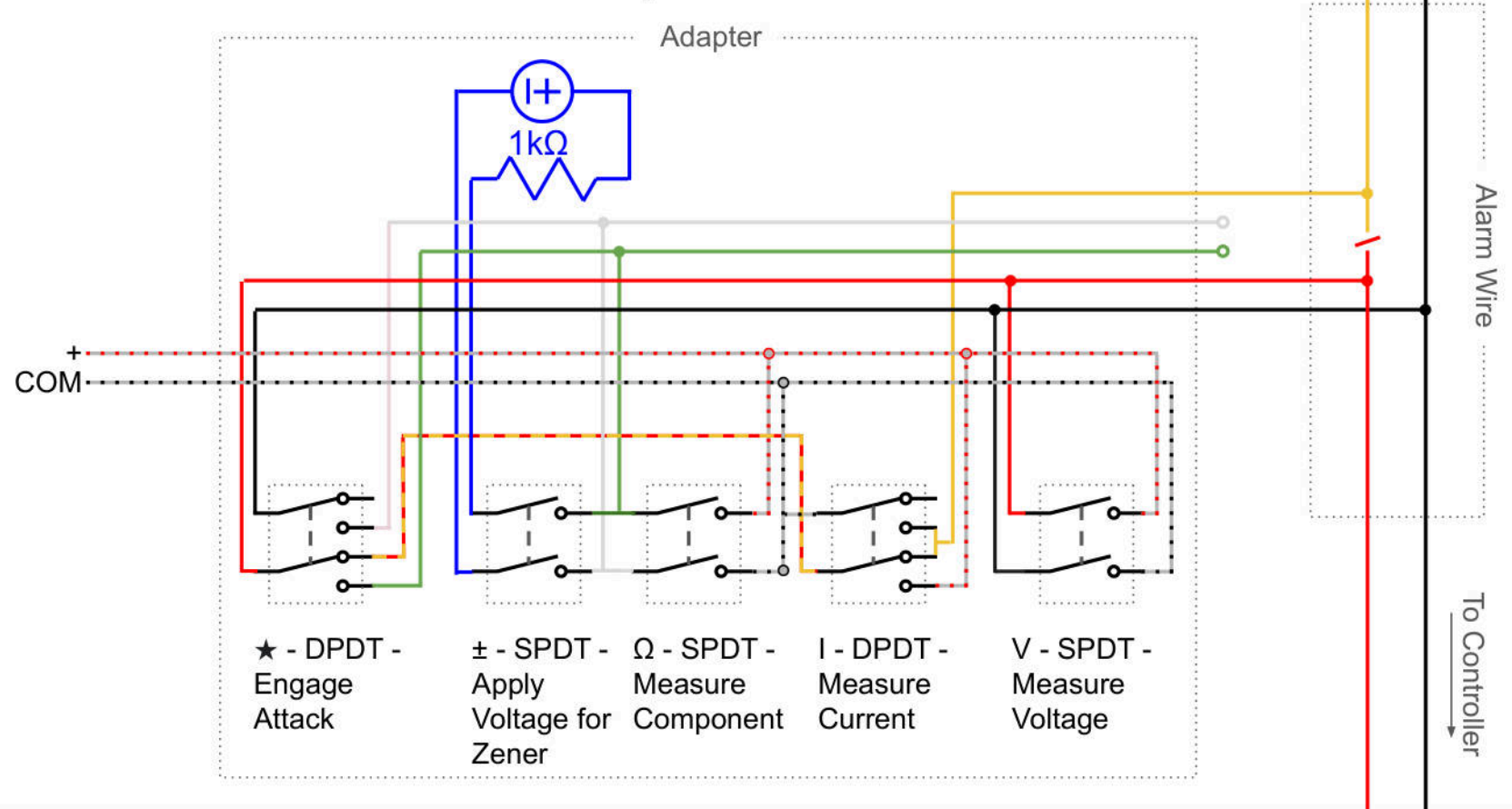
PILE ALCAINE

9V 无汞 DEC
MNT604 6LR61 2018

DT-830B
20170003



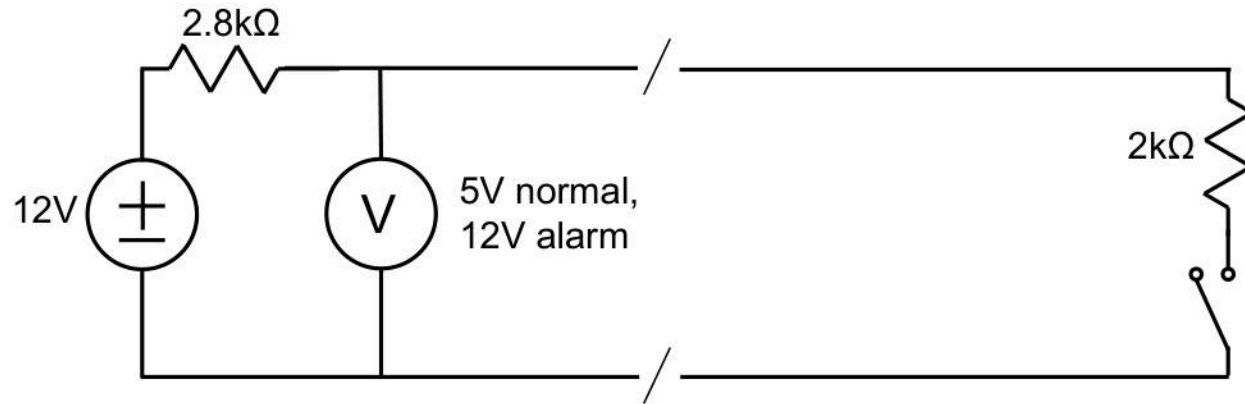
Schematic - Multimeter Adapter



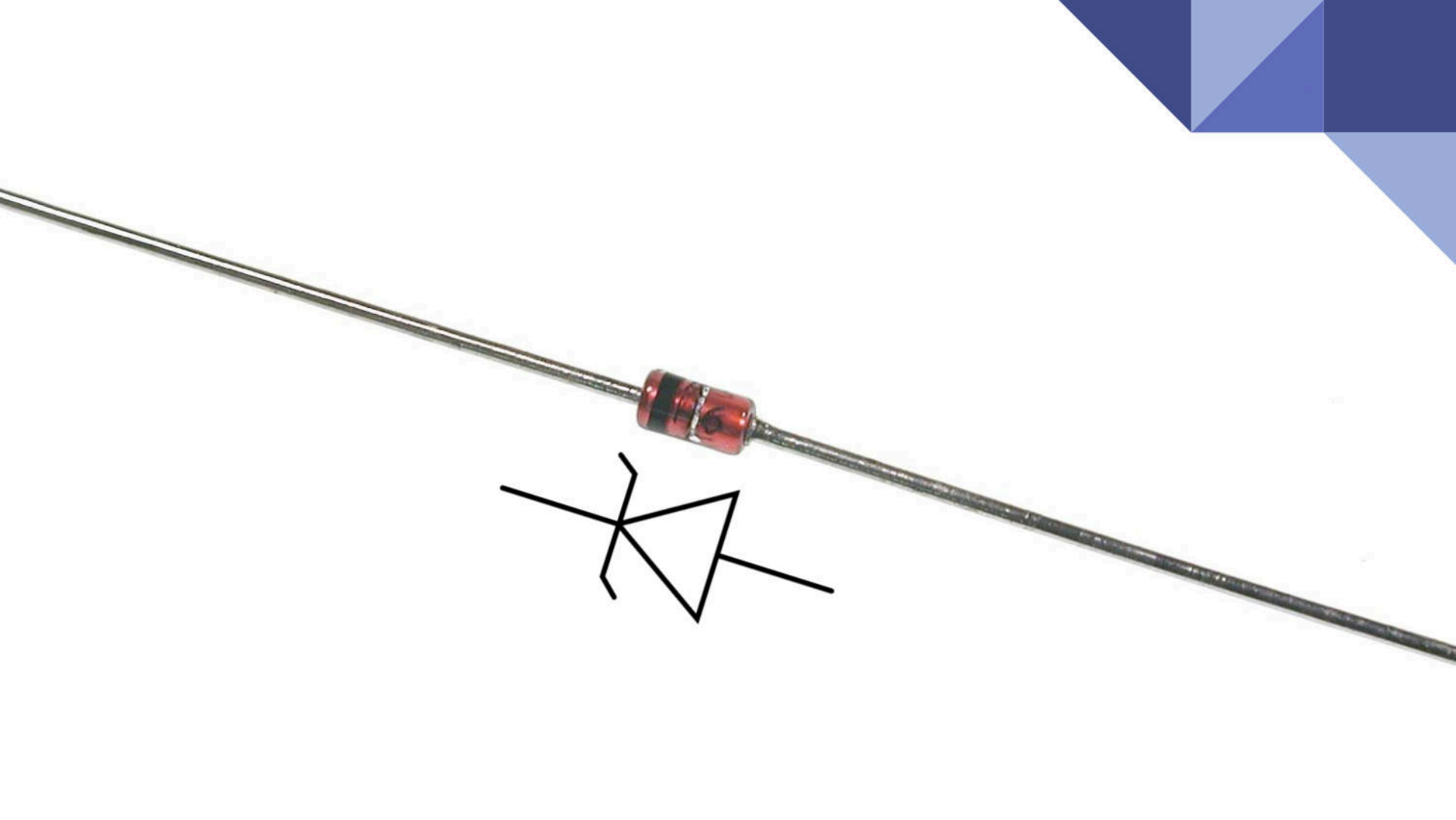


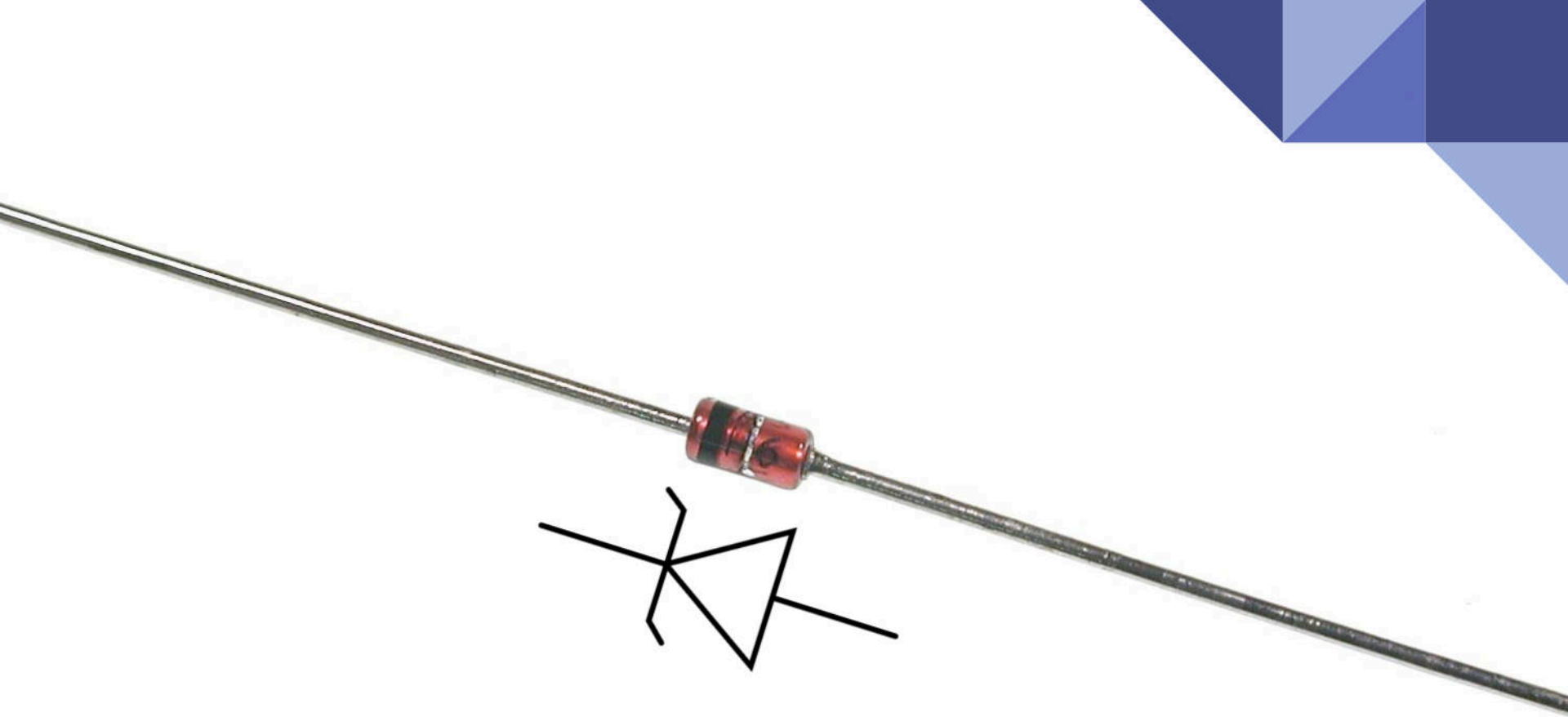
Can we do Better?

Can we do Better?









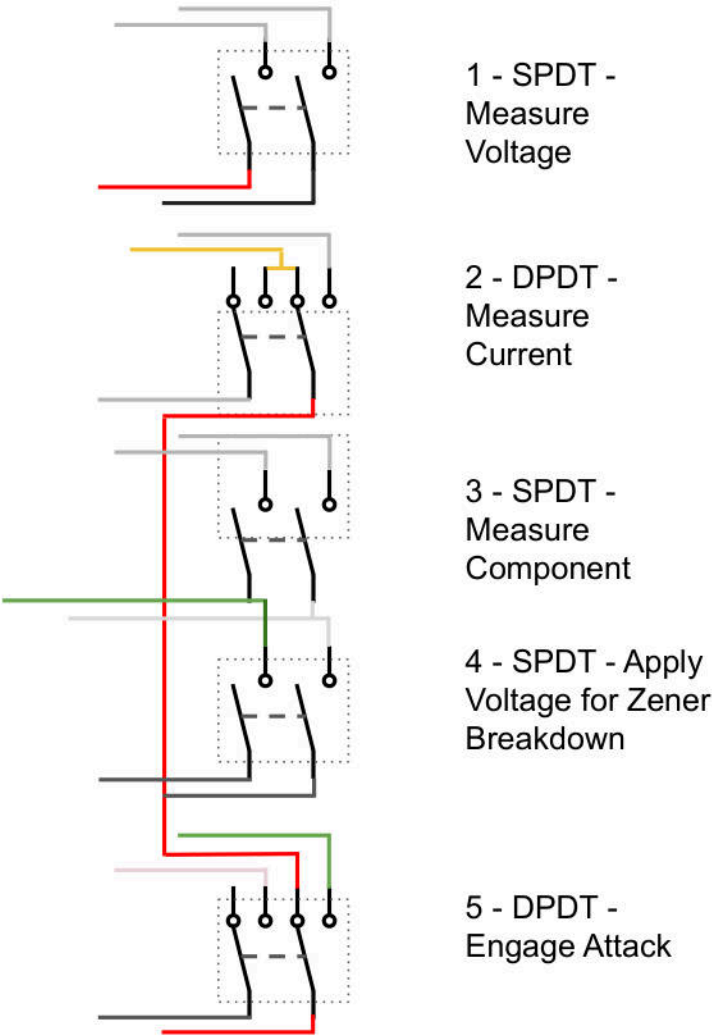
Demo →

Zener: effect changes with temperature; not well suited for outdoor applications if a high breakdown voltage (Avalanche effect dominates)

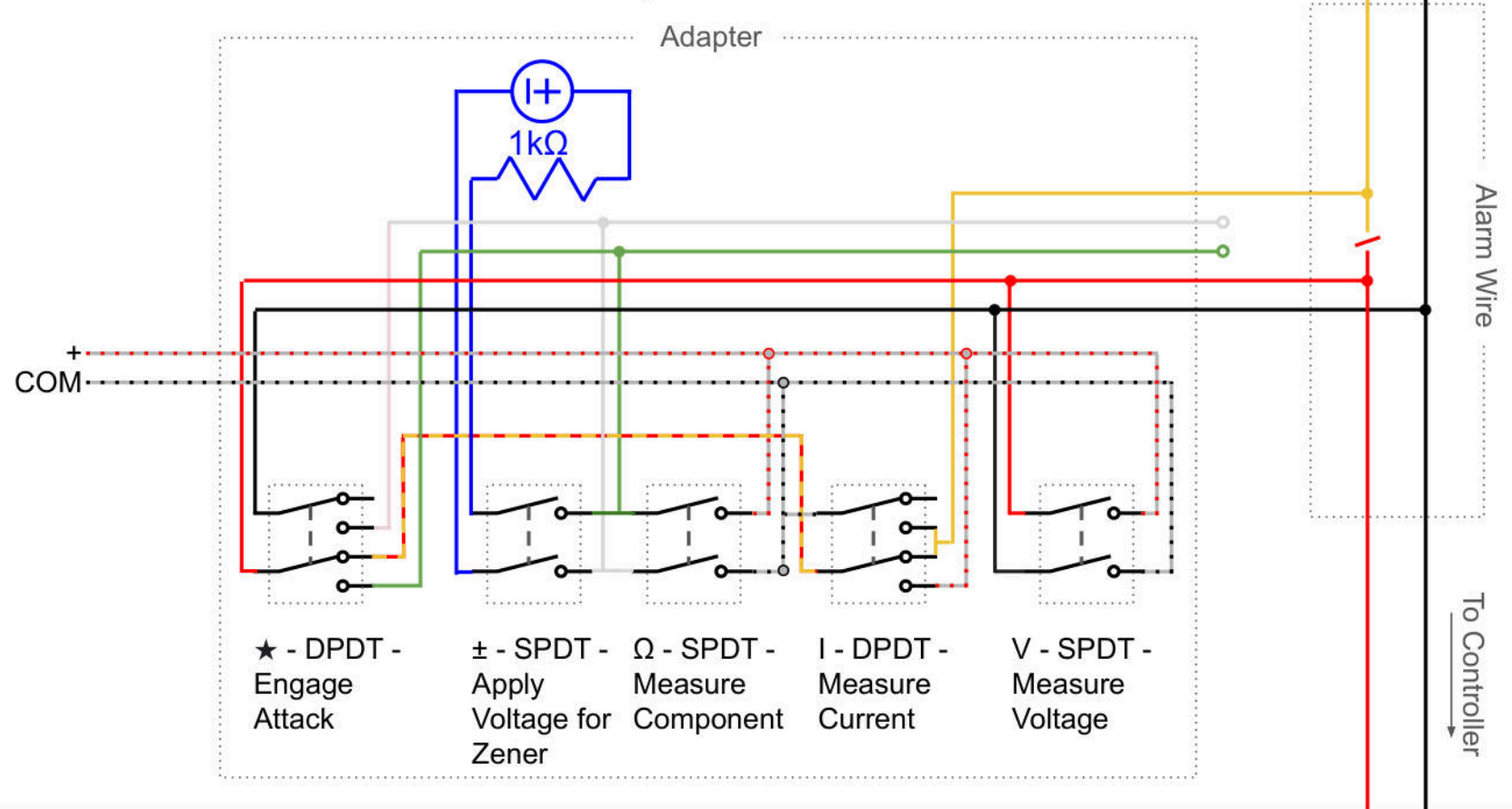
5.6V breakdown: Zener and Avalanche effect approximately equal, temperature coefficients cancel.

Low breakdown voltage: Zener effect dominates; much more rounded knee at breakdown. Less well suited - use resistive if possible.

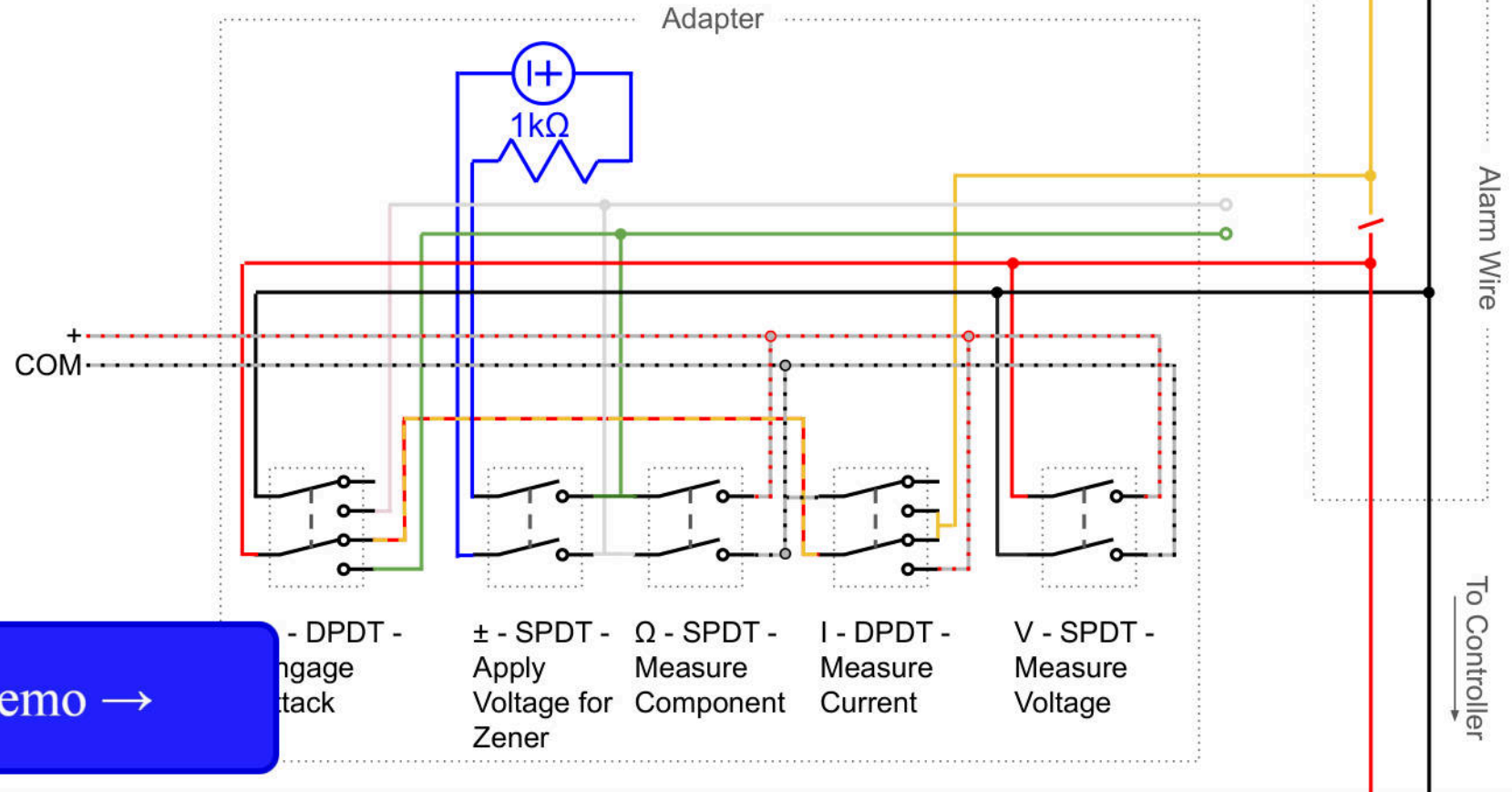
Schematic - Multimeter Adapter



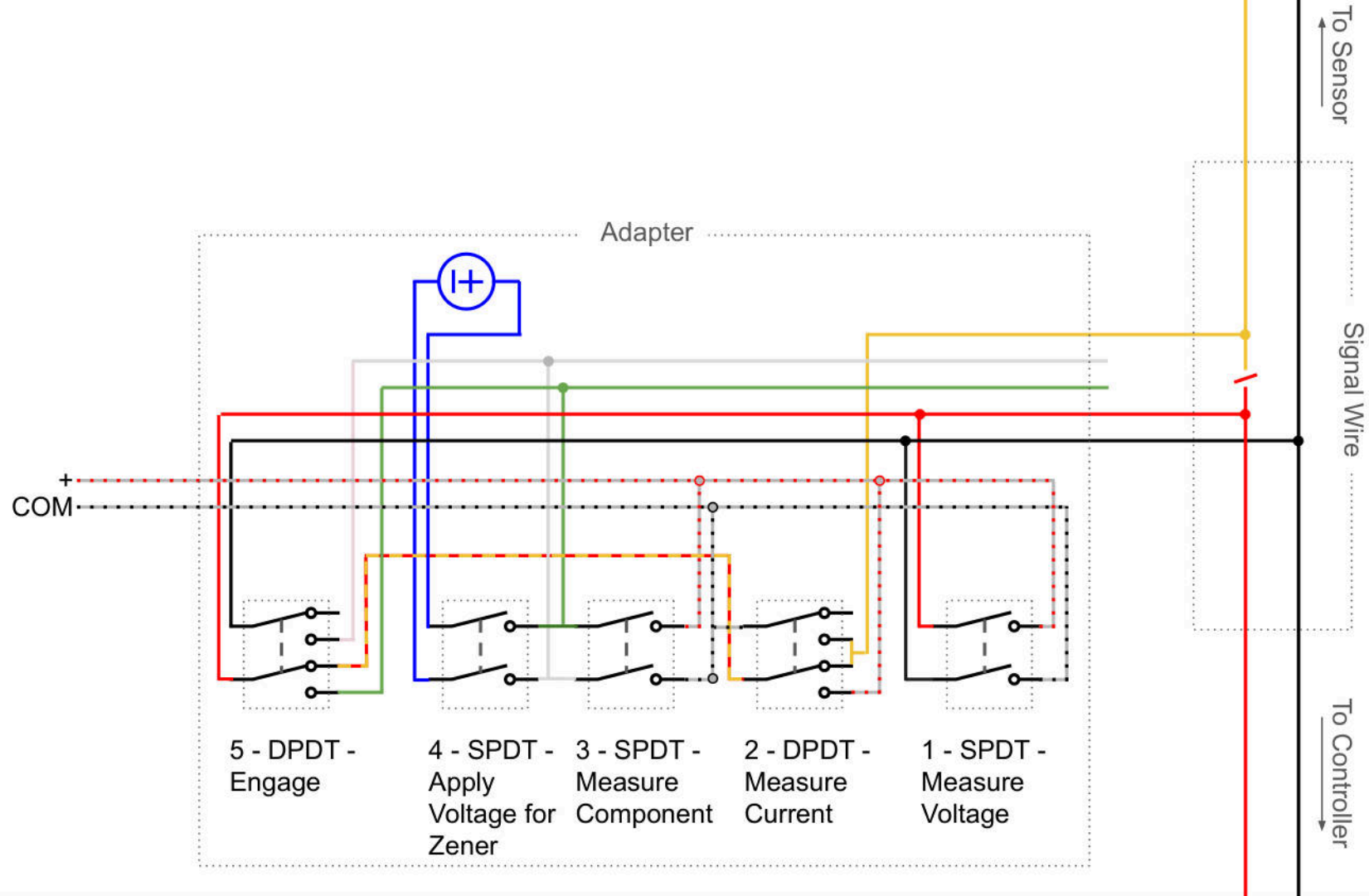
Schematic - Multimeter Adapter



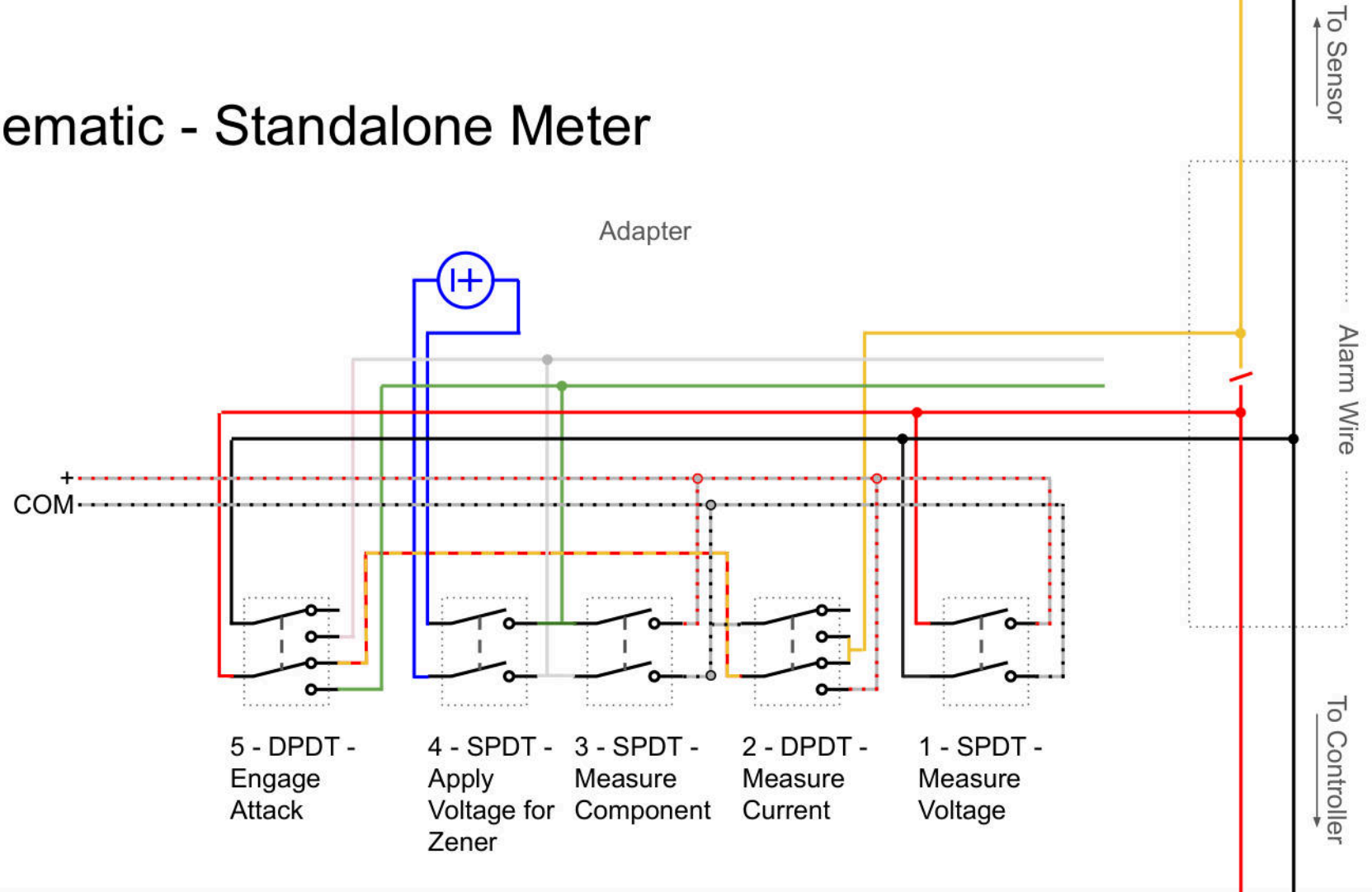
Schematic - Multimeter Adapter



Demo →



Schematic - Standalone Meter





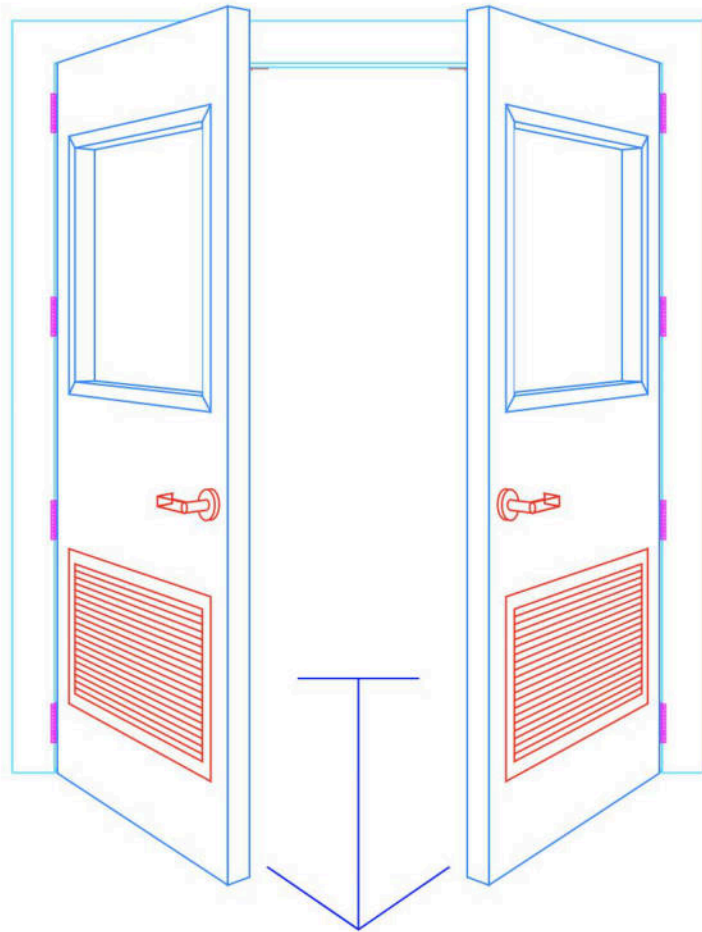
Defences



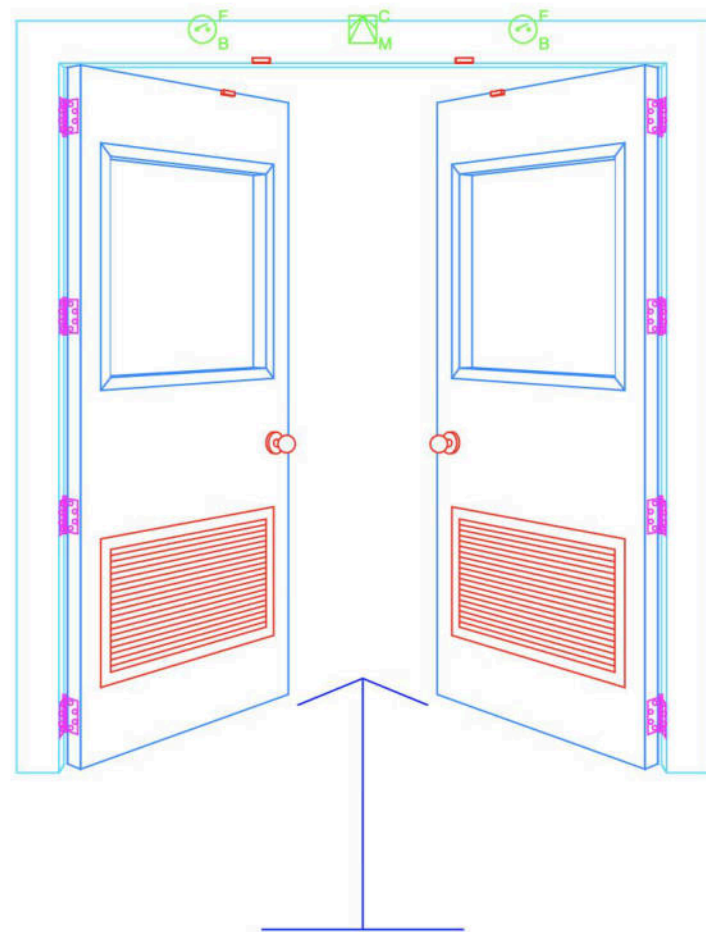


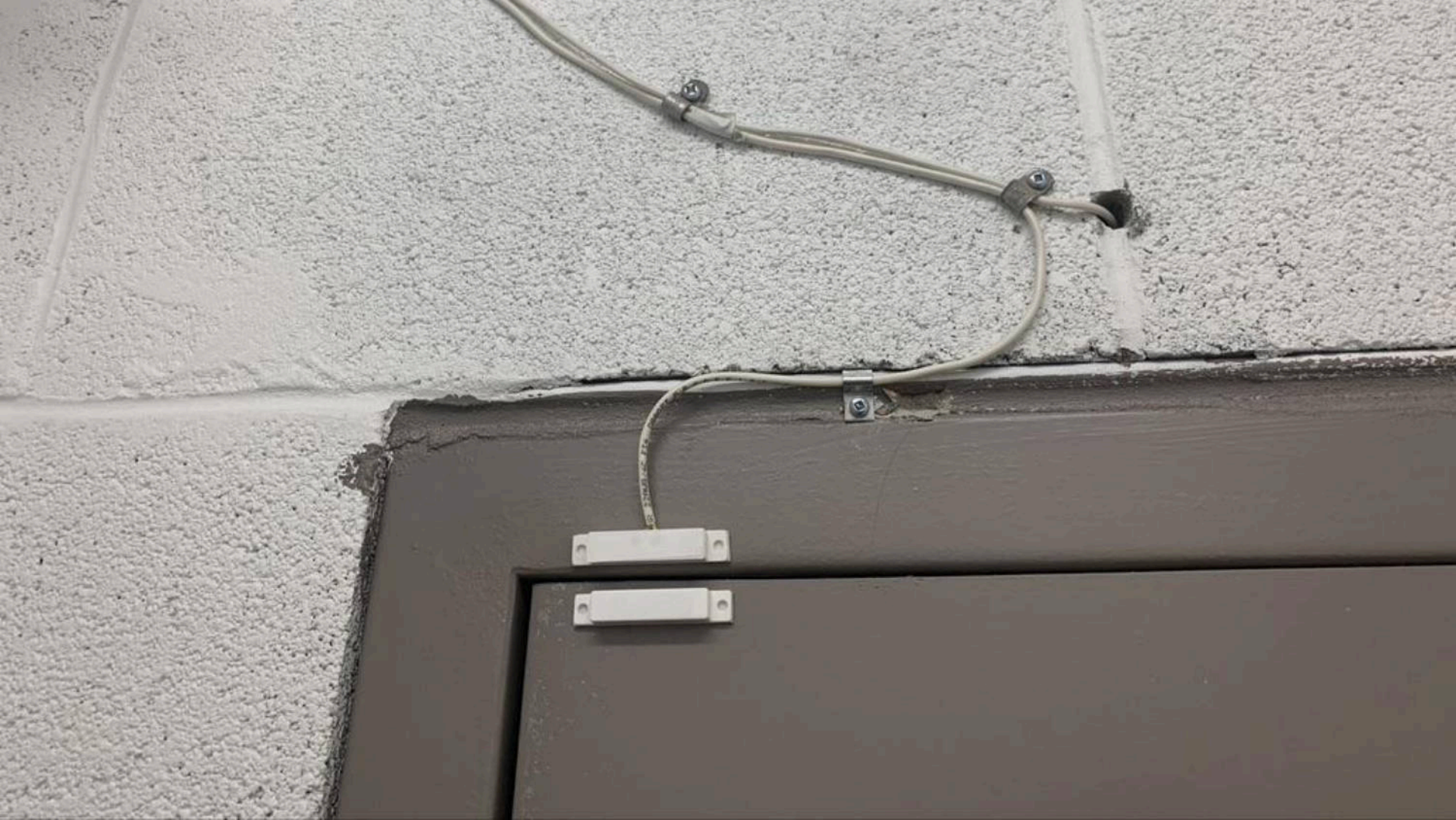


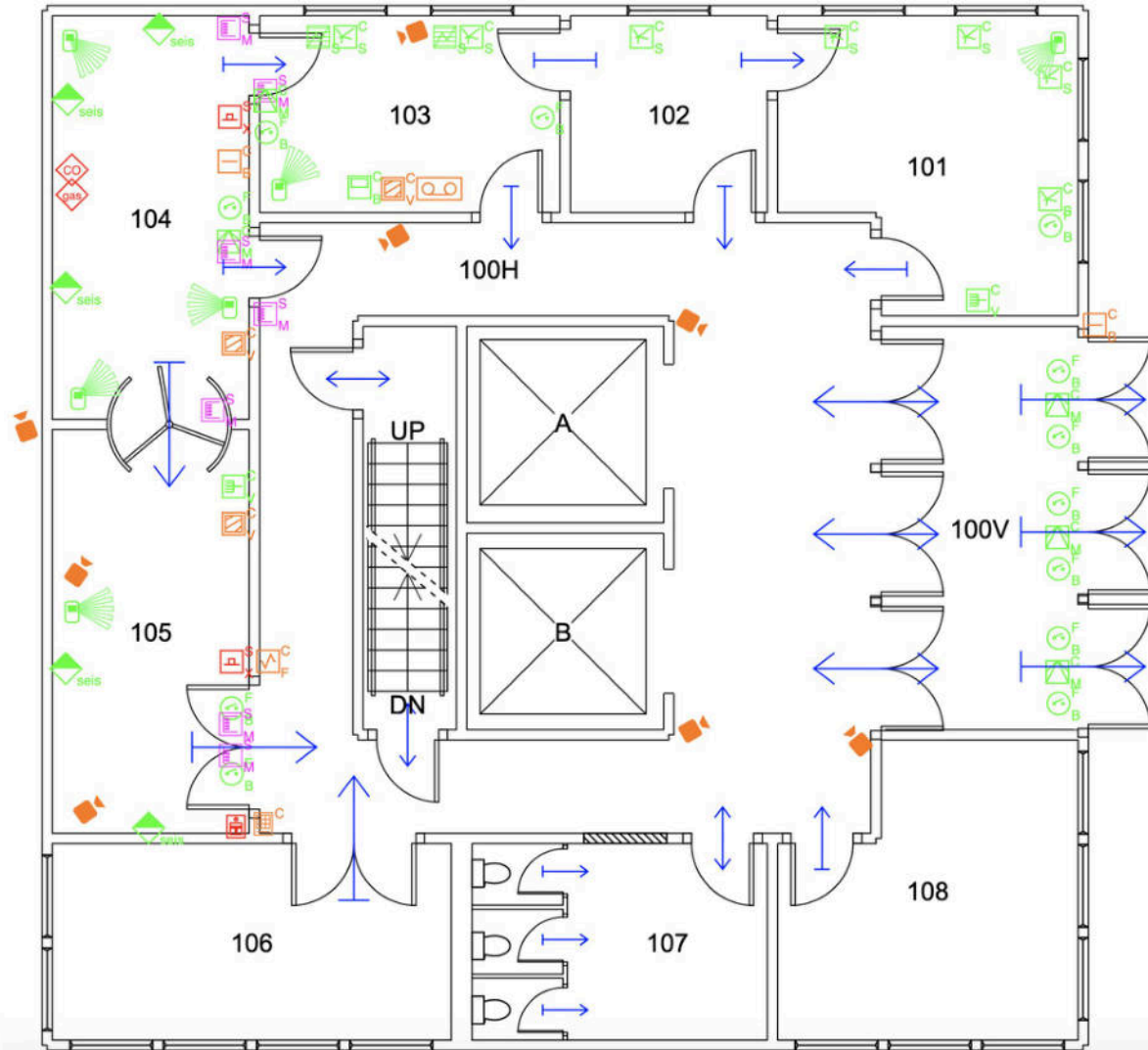
Side A



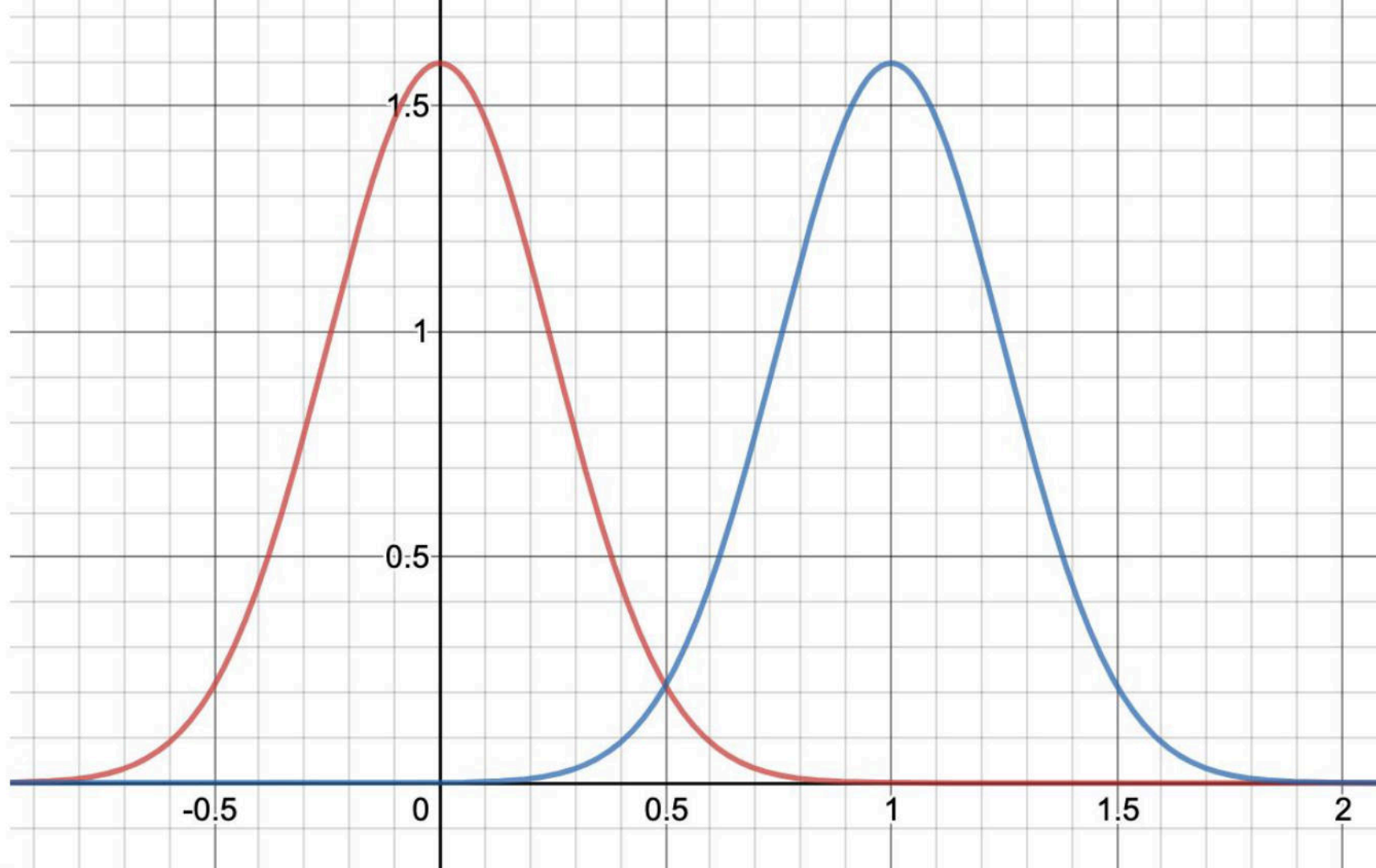
Side B



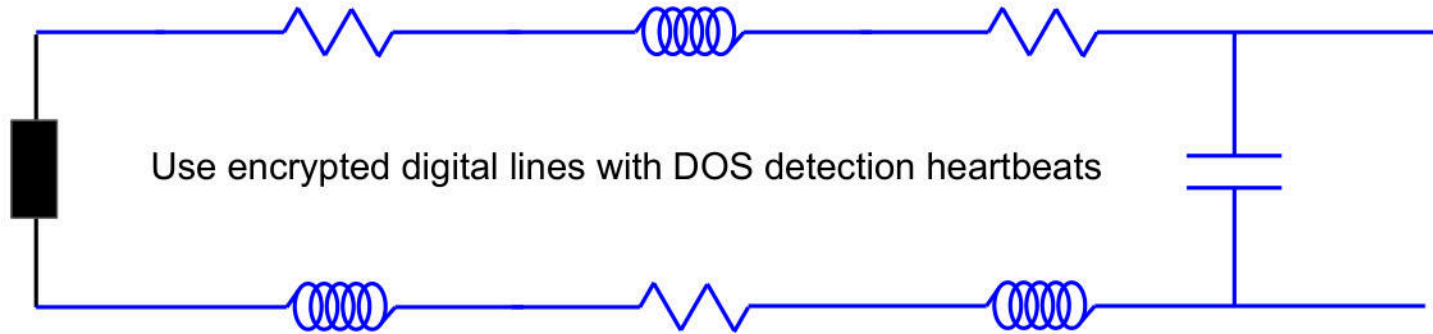








Encrypted Digital Lines





Operating Environment

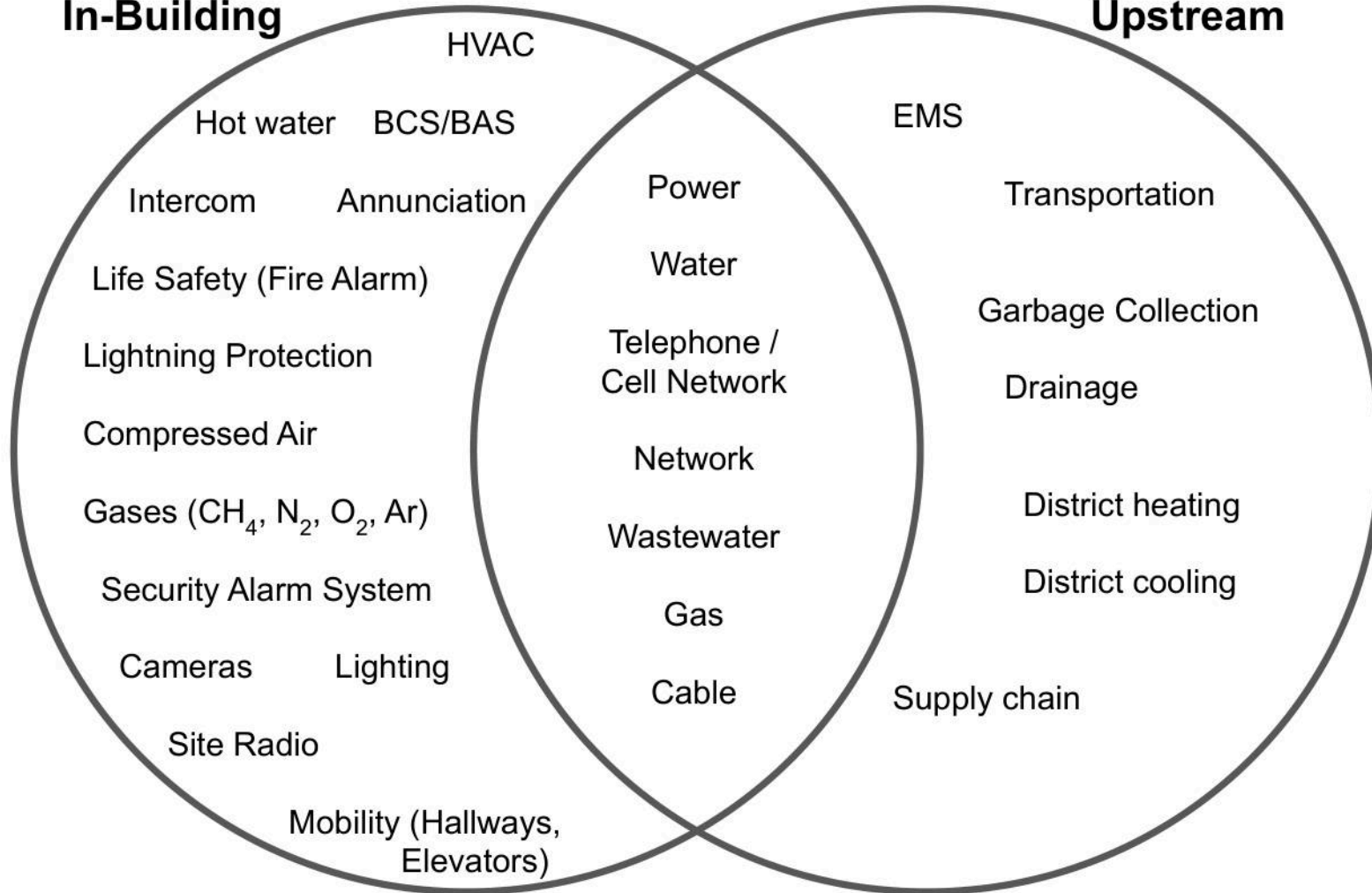
What do you rely on? What can change?





In-Building

Upstream



References

[1] B. Graydon, *Alarms and Access Controlled Doors: DEF CON Safe Mode - Lock Bypass Village*, August 6-9, 2020, Las Vegas, NV, USA. Available: <https://www.youtube.com/watch?v=hGUMUG9VLKU>

[2] N. Koch, *Inside Job: Exploiting Alarm Systems and the People Who Monitor Them: HOPE 2020*, July 25-August 2, 2020, New York, NY, USA. Available: https://www.youtube.com/watch?v=Rt_9dok3d_Q

[3] Physical Security, FM 19-30, Department of the Army, Washington, DC, USA, Mar. 1, 1979. [Online]. Available: https://www.jumpjet.info/Emergency-Preparedness/Disaster-Mitigation/Civil/Physical_Security.pdf

[4] R. Antunes, *Intruder Alarm Systems: The State of the Art*: Submitted to CEE'07 - 2nd International Conference on Electrical Engineering. Available: https://www.researchgate.net/profile/Rui-Azevedo-Antunes/publication/236982377_Intruder_Alarm_Systems_The_State_of_the_Art/links/5ec2c54492851c11a870c1ff/Intruder-Alarm-Systems-The-State-of-the-Art.pdf

[5] D. J. Brooks, *Intruder alarm systems: Is the security industry installing and maintaining alarm systems in compliance to Australian Standard AS2201?: Secur J* 24, 101–117 (2011). <https://doi.org/10.1057/sj.2009.12>

[6] B. A. Nadel, *Building Security*, New York, NY: McGraw Hill, 2004

[7] B. Graydon, *OSINT of Facilities by Physical Reconnaissance: HOPE 2020*, July 25-August 2, 2020, New York, NY, USA. Available: https://www.youtube.com/watch?v=BgovHNKh_fU

[8] K. Ng, *Bypass 101: DEF CON Safe Mode - Lock Bypass Village*, August 6-9, 2020, Las Vegas, NV, USA. Available: <https://www.youtube.com/watch?v=3yKZqiYGYnA>

[9] B. Phillips, *The Complete Book of Locks and Locksmithing, 7th ed.*, New York: McGraw-Hill Professional, 2017.



Questions?

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 [@access_ctrl](https://twitter.com/access_ctrl)

Go try it!

https://www.bypassvillage.org/games/alarm_wire/

Or just: [bypassvillage.org](https://www.bypassvillage.org)

Source:

https://github.com/bgraydon/alarm_wire

A huge thank you to Paul Robichaud, Karen Ng and Jenny & Bobby Graydon for their help in preparing this talk.